

YARG Contributions — Roadmap

Canonical roadmap for both repos in this project (`yarg-song-server` and the `yarg` client fork). Status as of 2026-09-05.

Two tracks run in parallel. The **server track** is the #1 priority and is sequential — each phase depends on the one before it. The **client track** is independent and can absorb spare cycles at any time.

What this project is, and what it is not

Written down because the roadmap drifted once already: upstream's issue tracker is full of interesting features, and a phase named after one of them starts reading like a commitment.

What we are building:

1. **A server/client pair that talks to YARG**, so a library lives in one place and every machine in the house plays from it — instead of a full copy of the library on every device.
2. **A way to manage that library without launching YARG at all.** Browse, search, inspect, see what is broken and why, from any device on the network with the game closed and no YARG install on that device. This is a first-class goal, not a side effect of having an API.
3. **Later: chart generation** — process audio into playable charts across all difficulties, with the charts distributable separately from the audio. Phase 5, and it depends on everything above.

What we are not building: features that live inside YARG and change how the game plays. Upstream has open requests and in-flight PRs in that space — search-and-queue from a phone, selecting a song in a running client. Those are **notes for a future conversation with the YARG team**, recorded in `UPSTREAM.md` , not deliverables here. A thing being adjacent to our API is not a reason to adopt it.

Phase 0 — Foundations ✅

Repos, remotes, mirrors and the format research that everything else is built on.

- [x] `fatalexception/yarg-song-server` and `fatalexception/yarg` created on Vault2 GitLab
- [x] Song-format research spec written (`docs/research/yarg-song-formats.md`)

- [x] Architecture decision recorded (`docs/ADR-001-server-architecture.md`)
- [x] Project instructions mirrored into `CLAUDE.md` so both machines pick them up
- [x] `yarg` imported from upstream server-side: 21 branches, 172 MB of repository and 180 MB of LFS objects, default branch set to `dev`
- [x] Public push mirrors to `github.com/Coffeehedake` configured and verified: `yarg-song-server` (GitLab mirror 10) and `yarg` (mirror 11), both one-way, all branches, divergent refs not kept. First sync 2026-09-05 16:44 ET, both `finished` with no error, and the GitHub side matches GitLab commit-for-commit.

Cloned 2026-09-06, ending a deferral this document argued for twice. It is on ENG-1 at `C:\dev\YARG - Open Source Contributions\yarg`, **625 MB** on disk with submodules and LFS — the earlier "~350 MB" estimate was low by most of a factor of two, which is the ordinary fate of a size guessed rather than measured. Branch `dev`, `origin` on Vault2 GitLab, `upstream` on GitHub, `YARG.Core` submodule initialised.

The deferral's original reason had already gone void: the folder used to sit in the Syncthing `dev-projects` mesh, so a clone here replicated to r7 and Vault2 as well. Syncthing was retired on 2026-09-06, so a clone now costs disk on one machine only — and the conclusion had been surviving on the weaker of its two arguments for a day before anything acted on that.

It was cloned to answer a question rather than to start work: **YARG.Core builds and tests with plain `dotnet` and no Unity install** (`netstandard2.1` library, `net10.0` tests), which is what makes `ADR-004` groundable in real types instead of guesses.

If it needs cloning again:

```
cd "C:\dev\YARG - Open Source Contributions"
git clone https://gitlab.badassium.com/fatalexception/yarg.git yarg
cd yarg
git remote add upstream https://github.com/YARC-Official/YARG.git
git submodule update --init --recursive
git lfs pull
```

Exit criterion: both repos exist, both push to Vault2, both mirror to GitHub, and the format spec is committed. **Met 2026-09-05.**

`Coffeehedake/yarg` is a real GitHub *fork* of `YARC-Official/YARG`, not a plain repository, because that is the only shape GitHub accepts a pull request to upstream from.

Correction — LFS does propagate. An earlier revision of this roadmap claimed GitLab push mirroring drops LFS objects, and that the fork relationship was what made upstream's ~180 MB resolve. That was asserted from priors, not measured, and it is wrong.

The measurement: a throwaway GitLab project (**not** a fork, so no parent LFS storage to borrow from) with `*.png filter=lfs`, one fresh 1 MB object, push-mirrored to a throwaway **non-fork** GitHub repo. GitHub's LFS batch API returned a download action with no error, and the object downloaded from `github-cloud.githubusercontent.com` at exactly 1,048,576 bytes with a SHA-256 matching the OID. GitLab 18.11.3, git-lfs 3.7.1. Both throwaway repos were deleted afterwards.

So no special handling is needed when client work adds one of the five patterns YARG's `.gitattributes` tracks — `*.png`, `*.exr`, `*.jpg`, `*.fbx`, `*.ttf`. Which is just as well: almost any UI work in Phase 3 adds a `.png`.

The caveat that does survive: the mirror force-overwrites and the GitHub side is a fork, so anything done directly on GitHub is clobbered on the next sync. To open an upstream PR, create the branch on GitLab, let it mirror, open the PR from the mirrored branch, and leave it alone on GitHub after that.

Phase 1 — `yargsong`: the Go format library ✅

The server is worthless until Go can read and write what YARG reads and writes. This phase is pure library work with no network surface, which makes it the easiest phase to test exhaustively.

Build order (each step is independently testable):

1. **`song.ini` reader** — ✅ done (`internal/songini`). ~130 recognised keys with their types, and a deliberately lenient reader: UTF-8/UTF-16 BOMs, Latin-1 fallback so accented artist names survive, `[song]`/`[Song]`/no-header variance, trailing junk after the section bracket, values containing `=`, and malformed lines skipped rather than fatal. Two behaviours were checked against upstream rather than guessed: **duplicate keys — last wins** (upstream stores modifiers by plain dictionary assignment), and **keys and section names are lowercased** before lookup. One thing is still assumed and flagged in the source: the exact whitespace and multiple-`=` handling inside `YARGTextReader.ExtractModifierName`, which has not been read. The writer is step 5 below.
2. **`.sng` reader** — ✅ done (`internal/sng`). Implemented as an `fs.FS`, including synthesised directories, so the folder scanner and the `.sng` scanner share exactly one code path; `fstest.TestFS` enforces that. The mask-origin question the research doc flagged as unconfirmed is now settled from `SngFileStream.cs`: **the XOR index is the byte offset within each contained file, restarting at 0 per file**, not the absolute offset in the `.sng`. Upstream reaches the same result only because it decrypts whole 1 MB buffers and 1 MB is a multiple of the 256-byte table; tracking the real offset is equivalent and survives arbitrary read sizes. Malformed input is rejected with an error rather than a panic, which is tested.

Validated against the reference encoder, 2026-09-05. SngCli v0.3.0 (MIT, from `mdsitton/SngFileFormat` — the tool that defines the format) encoded a song folder we wrote, and our reader read the result: same chart bytes, same SHA-1, same metadata, and chunked streaming matching a whole-file read. The archive is committed at `internal/sng/testdata/reference-sngcli-v0.3.0.sng` so this stays a regression test rather than a thing that was true once. Until this, every `.sng` the reader had seen was produced by an encoder written from the same understanding as the reader — which proves only that the two agree with each other.

Two things the real file taught us that synthetic input could not:

- **Duplicate `song.ini` keys: last wins — confirmed independently.** A probe archive with `name = FIRST VALUE` … `name = SECOND VALUE` round-tripped through SngCli as `SECOND VALUE`, matching what we had concluded from reading `IniModifierCollection.cs`.
- **A file's extension can lie about its container.** SngCli emits audio under a `.mp3` name regardless of source format: our `song.wav` came back as `song.mp3`, byte-identical, RIFF header intact. We classify stems by name, as YARG does, so the scan is unaffected — but anything that later *decodes* audio must sniff the container, and our own writer must not reproduce this. There is a test pinning the observation.

- **Scanner** — ✅ done (`internal/scan`, `internal/catalog`). Walks a folder or a `.sng` through the same `fs.FS`, applies the chart-file priority order, classifies stems (all 14 standard, 5 clean and 4 explicit variants), resolves album art with the `cover` key overriding `album.<ext>`, finds background/video/preview, and emits the v1 catalog schema. Unrecognised files are carried in `assets.other` and every `song.ini` key is preserved in `raw_metadata`, so parse-tuning keys we do not model still survive a repack — losing those would change how a chart plays.

A test packs the same content both ways and asserts the folder and the `.sng` produce the same chart hash, metadata and parts. That is the ADR-001 "one code path" claim being enforced rather than merely intended.

Unknown intensity is `-1`, never `0`, for both an explicit `-1` and an absent key — "the charter rated this trivially easy" and "the chart did not say" must not collapse into the same value. Two `diff_*` keys (`diff_drums_real_ps`, `diff_keys_real_ps`) are deliberately left unmapped rather than guessed; the source says why.

`yarg-song-server scan <path>` walks a library and prints the catalog as JSON, so the parsers can be pointed at a real collection before anything sits behind an HTTP API. 4. **Identity** — ✅ done. `SHA1(chart file bytes)`, matching YARG's `HashWrapper`, plus a `package_hash` of our own (SHA-256 over the sorted `name:sha256` pairs) for same-chart-different-audio cases. Note the folder and `.sng` forms of one song have the same chart hash but *different* package hashes, because the folder carries `song.ini` as a file and the `.sng` carries it in the header — that is correct, and the test asserts it rather than papering over it. 5. **`.sng` writer** — ✅ done (`internal/sng/write.go`, `scan.PackDir`, `yarg-song-server pack <folder> <out.sng>`). Validated three ways, weakest first:

- **Our own round-trip** — proves only that our reader and writer agree.
- **The reference decoder.** `SngCli decode` extracted our archive with zero errors, and `notes.chart`, `album.png` and the audio all came back byte-identical to the source folder. Our archive of the same song is the same size as SngCli's, 8,806 bytes.
- **The client.** YARG v0.15.0 scanned 15 archives written by us and accepted **14**, reading every title correctly out of our metadata section — including Latin-1, UTF-16LE, a duplicate key resolving to `SECOND`, and an unknown key surviving the repack.

The one rejection was `No notes found`, and a **control settles it**: SngCli's own archive of the same source folder, scanned alongside ours in the same pass, was rejected with the identical error. The fixture's hand-written chart has no note events. As far as the client is concerned our writer is indistinguishable from the reference encoder.

Deliberate differences from the reference encoder: we do **not** rename audio to `.mp3` (SngCli does this regardless of the source container), and we refuse `song.ini` as a contained file rather than letting it disagree with the metadata section. Filenames are lowercased and collisions after lowercasing are refused, metadata keys are emitted lowercase because YARG matches `.sng` keys against a lowercase table without normalising them, and the header size is asserted against what was actually written — a mismatch there would silently corrupt every file offset. 6. **Chart preparsers** — ✅ done (`internal/chart`). Determines which parts and difficulties a chart actually contains, for both `notes.mid` and `notes.chart`. No timing, no sustains, no HOPO inference — the browse UI needs an instrument grid, and the client already has YARG.Core for everything else.

Built from documentation this time, not from source: TheNathannator's Guitar Game Chart Formats, the RBN/C3 documentation, FireFox2000000's `.chart` spec and the Elite Drums spec. The study is in `docs/research/chart-preparsing.md` with a citation per claim.

The governing rule is never to range-test a difficulty block. Every instrument family has non-playable numbers inside its own blocks, and a range test turns each into a phantom difficulty. Sixteen tests exist mostly to pin the traps the documentation names:

- force-strum and HOPO markers sit inside the block and are not notes;
- five-fret note 59 is a left-hand *animation* note unless an `ENHANCED_OPENS` event says otherwise — counting it unconditionally invents an Easy difficulty on a large share of Rock Band-derived charts;
- every Pro Keys track is *required* to carry a range-shift marker, so a test not strictly inside 48–72 reports even empty tracks as present;
- the Pro Keys animation tracks use an identical note range and differ only by name, so track names are matched whole-string, never as substrings;
- Elite Drums' modifier octave carries a disco-flip marker that exists for downcharting and can sit on a difficulty with no gems;
- a lone `HARM1` is the lead line, not a harmony arrangement.

Drum type is a heuristic because it has to be — 4-lane, Pro and 5-lane share one track — with `song.ini`'s `pro_drums` / `five_lane_drums` overriding, and both set at once recorded as the documented invalid state rather than silently resolved. Elite Drums downcharting is honoured: a song with only `PART ELITE_DRUMS` reports 4-lane, Pro and 5-lane as `derived`, because the client shows them.

The SMF reader is hand-rolled rather than a library, deliberately. Charts violate the MIDI spec in two documented ways — running status not reset after SysEx or meta events, and `0xFF` bytes inside SysEx — and a strict parser rejects files YARG plays fine.

Cross-checked against the oracle: the corpus case whose `notes.mid` YARG rejected with "No notes found" now reports zero parts from our preparser too. Same conclusion, independent route.

Exit criterion: a CLI that scans a real song library, emits a catalog, repacks to `.sng`, and YARG scans the repacked output with identical metadata and an identical hash. **Met 2026-09-05.**

`yarg-song-server scan <path>` and `yarg-song-server pack <folder> <out.sng>` do the first two. For the third: `SngCli` decoded our archives with every file byte-identical to the source, and YARG accepted 14 of the 15 archives we wrote — the one rejection being a fixture whose chart has no note events, proven by a control in which SngCli's own archive of the same folder was rejected identically.

And the parts we report now agree with the client's own verdict: on the corpus (22 cases then, 23 since archive ingest landed, 26 since the bare-CR cases), **every song YARG rejects is one this scanner independently flags**, by three routes — no parts detected, `no_audio`, and `ultrastar_no_title`.

A fourth code, `metadata_unreachable`, is **not** a refusal class and must not be counted as one. YARG plays those songs; it just cannot see one field the charter wrote, because a bare CR or a key its own table cannot hold put that text out of reach. It is reported so a missing field has an explanation. See the tenth oracle run in `docs/TEST-CORPUS.md`.

Explicitly out of scope, permanently: CON/mogg decryption, `songcache.bin` generation, `.milo_xbox` / `.png_xbox` decoding, `.yarground` inspection, full MIDI chart semantics.

Phase 2 — The server, and a sync client that needs no client changes

Two deliverables. The sync client is what makes the server *useful* before any client fork work exists, and it is the proof that the server is correct.

2a — `yarg-song-server`

- [x] **HTTP API** — `internal/httpapi`, documented in `docs/API.md`. Browse and search over the 12 attributes YARG itself sorts by, `GET /song/{chart_hash}.sng`, and a bulk `POST /api/v1/have` that takes the hashes a client holds and answers what it is missing.
- [x] **The index and the store** — `internal/library`, in memory and rebuilt at start; `internal/packcache` packs a loose folder to `.sng` on demand and caches it by package hash. Both decisions and their costs are in `docs/ADR-002-v1-store.md`.
- [x] **Sort parity with the client** — `internal/sortkey` reproduces YARG.Core's `SortString` (rich-text stripping, diacritic folding, whitespace collapse, article removal, character grouping, UTF-16 ordinal comparison) and `internal/library` orders by upstream's own comparer chains. Without this a browse list is internally consistent and unlike anything the player sees in the game.
- [x] **Ingest: loose folders, `.sng`, and `.zip` / `.7z` of a loose folder** — done 2026-09-06. `internal/scan/source.go`. `.zip` uses the standard library; `.7z` took a dependency, recorded with its measured cost in `ADR-003` (+1.62 MB, 3 → 71 compiled packages, no cloud stack despite what `go list -m all` implies).

  `.rar` **joined on 2026-11-01**, as a full container rather than unpack-only, via `github.com/nwaples/rardecode/v2` — +0.43 MB and **one** compiled package, because it is pure Go with no dependencies of its own. The decision hinged on a measurement, not on the format: reading a `.rar` entry OUT OF ORDER costs what reading the first one costs (<1 ms), and enumerating a 6 GB archive takes 33 ms, where a solid `.7z` costs about three minutes per out-of-order entry. Only the FIRST volume of a multi-volume set is an entry point. Numbers, caveats and the absent fixture are in `ADR-007`.

  It shipped unable to unpack a `.rar`, and the tests could not see it. RAR marks a directory with `FileHeader.IsDir` and no trailing slash, where `archive/zip` and sevenzip both append one — 45 directory entries in Modern Hits, none slashed, against 6 of 6 in a `.7z`. So every song folder read as a zero-byte file. READING tolerated that (`archiveFS.record` promotes a path to a directory as soon as a child names it), which is why all 44 songs were found, every one packed, and the gated live test passed; WRITING did not, because `writeOne` uses `O_EXCL` on purpose. Fixed at the reader boundary in `storedName` (`fab8c90`), and the live test now EXTRACTS as well as reads. The rule it earned: a forgiving reader hides a strict writer, so a new container is exercised through extraction and not only through the `fs.FS`.

  No adapter was needed and no scanning logic was duplicated. `archive/zip`'s Reader and sevenzip's Reader both already implement `fs.FS`, and the scanner has taken an `fs.FS` since Phase 1, so an archived song is read by exactly the code that reads a loose folder. `PackDir` became a one-line wrapper over a new `PackFS`.

  The property that matters: **a song packs to the same bytes whether it is loose, zipped or 7z'd**, so restructuring a library does not make every client re-download every song. `TestAllThreeContainersProduceTheSameArchive` asserts it directly, and the corpus now carries a zipped case (`23-zipped`) so the oracle exercises it end to end.

RB packages are refused with a reason, not ignored — `.con`, `_rb3con`, `.pkg`, `.xex`, matched by SUFFIX because `_rb3con` files usually have no extension at all. An archive holding several songs raises `ErrTooManySongs` rather than publishing one and silently dropping the rest.

Hardened against hostile archives, by probing rather than by reasoning — a throwaway test printed what actually happens for traversal entries, corrupt files, empty archives and odd separators, and only then were assertions written. Path traversal is dropped by `fs.ValidPath` before it can reach a served `.sng`; a corrupt archive is reported and the walk continues.

The probe found a real defect: a zip written with **backslash** separators (`Song\song.ini`, as some Windows tools produce) surfaced a directory with nothing readable under it, resolved to `ErrNoChart`, and was **silently ignored** — a legitimate song vanishing from a library with no message, the same failure this project had already rejected for `_rb3con`. Such an archive now reports `ErrUnreadableArchive`. `internal/scan/hostile_test.go`; the shapes measured are tabulated in `TEST-CORPUS` and the reasoning is in `ADR-003`. - [x] **Config file + sane defaults** — `internal/config`. `key = value` with `#` comments, the same names as the flags, precedence defaults < file < flags actually typed, and `--write-config` to print a commented example. An unknown setting is an error, not a warning. A file NAMED on the command line and missing is fatal; the conventional `./yarg-song-server.conf` simply not existing is the normal first run and is silent. - [x] **CI** — `.gitlab-ci.yml`. `gofmt`, `go vet`, the suite, the suite again with `-race`, all six release targets, and an assertion that the Dockerfile's Go is not older than `go.mod` requires. See below for what this replaced. - [x] **A bound and an eviction policy for the pack cache** — done. `pack_cache_max`, defaulting to **2 GiB** rather than to unbounded, with LRU eviction. The number that decided the default was measured on the vault2 deployment: 225,406 bytes of loose library produced 229,515 bytes of cache across 22 archives, **a ratio of 1.02** — so an unbounded cache over a loose library eventually needs a second copy of that library on the data disk. On the Pi target, with the library on external storage and `-data` on the SD card, that fills the card. Eviction costs a re-pack and never data, because the archive is rebuilt byte-identically and its package hash comes from the content. **That last sentence was written on 2026-09-05 and was untrue until 2026-09-06** - packing drew a random mask, so a re-pack produced a different archive; see the determinism entry below. Two smaller leaks went with it: the per-key lock map (now 256 fixed shards, bounded by construction) and orphaned `.partial` files from a crash mid-pack (now swept at start). `internal/packcache` had no tests at all; it now has six, each red-proofed. - [x] **The multi-arch container image** — `container-image` in `.gitlab-ci.yml`, pushing `linux/amd64` + `linux/arm64` to `registry.badassium.com/fatalexception/yarg-song-server`.

And it needs no emulation, which is the part worth remembering. The earlier plan here was to install qemu/binfmt on Vault2. That plan was wrong. Go cross-compiles natively, and the Dockerfile already pins its build stage with `FROM --platform=$BUILDPLATFORM`, so the toolchain runs at the host's own architecture and Go emits the arm64 binary itself. Emulation only enters if the build stage is pulled *for* the target platform, which that directive prevents. **Nothing on the Vault2 host was changed, and nothing should be.**

That correction came from the ShopStack session, who also established that `juniper-pi-deploy` — the "Pi framework" this was going to be cloned from — builds bootable SD-card images and publishes no container images at all. Same word, different problem.

Two things the job does that are not obvious:

- **It creates a `docker-container` buildx builder.** The default builder uses the `docker` driver, which cannot build more than one platform and cannot produce a manifest list at all. Without this step buildx fails with a message about the driver rather than anything that mentions multi-arch.

- **It verifies the result rather than trusting the exit code.** `ci/verify-multiarch.sh` asserts the manifest covers both platforms *and* reads the ELF machine type of the binary inside the arm64 image. An amd64 binary under an arm64 manifest entry passes every check that only reads exit codes, and then fails to start on the Pi. The check is structural rather than execution- based on purpose: running the arm64 binary in CI would need the emulation this project deliberately does not have.

That structural check was doing its job, but it is not the same as an execution, and this document said so for weeks. On 2026-09-06 the gap was closed on real hardware instead of in CI — a Raspberry Pi 4 ran both the arm64 binary and the published arm64 image. CI's job is unchanged: catch a mislabelled manifest early. Proving it *runs* stays a hardware exercise.

What CI replaced. This project had no `.gitlab-ci.yml`, so GitLab fell through to Auto DevOps. Pipeline #2216 — the only pipeline this project has ever run — failed three jobs with exit 127 trying to execute `/build/build.sh`, `/bin/herokuish` and `lsif-go`. Those are container-executor assumptions, and the runner is a **shell** executor on Vault2, which has none of them. A red pipeline nobody believes is worse than no pipeline, so the CI here brings its own pinned Go toolchain (SHA-256 verified before use, cached between runs) rather than depending on whatever happens to be installed on the host.

Measured end to end on 2026-09-05, against the 22-case corpus: the server indexed all 22 in 15 ms, `POST /have` from an empty client reported 22 missing, all 22 downloaded through `GET /song/{hash}.sng`, and re-scanning the downloaded archives gave 22 songs and 0 failures.

Then the oracle, which is the only test that means anything here: **YARG v0.15.0 scanned the 22 archives the server produced and accepted 20.** The two it refused were the two corpus cases built to be refused, and our own scanner flags both independently — `No notes found` against a preparser reporting no parts at all, and `No audio accompanying the chart file` against our `no_audio` issue. The standard holds on the served archives as well as on the folders.

One thing that run turned up, worth knowing before Phase 2b. Repacking a loose UltraStar folder whose `notes.txt` has no `#TITLE` makes YARG *accept* a song it refuses as a folder. As a folder the title has to come from the chart, and a missing `#TITLE` is fatal — "Name metadata not provided". Packed, the name lives in the `.sng` metadata section, which the packer fills from `song.ini`, so the song has a title and plays. Our scanner agrees with the client in both forms (`ultrastar_no_title` on the folder, no issue on the archive), so nothing here is broken — but it means "the server serves exactly what the folder was" is not quite true for this one format, and a sync client will show a song the player could not previously play.

2b — sync client

A small companion binary that pulls the server's library into an ordinary local songs folder. Unmodified YARG then just sees files. This ships real value in days, with zero risk to the client, and exercises the whole server end-to-end.

- [x] **`yarg-sync`** — `internal/syncclient` and `cmd/yarg-sync`, documented in `docs/SYNC-CLIENT.md`. Writes `<chart_hash>.sng` and nothing else, verifies every download by re-deriving identity from the bytes it received, resolves a shared chart hash deterministically, and leaves everything it did not write strictly alone — including under `-prune`, which is off by default. Built for all six release platforms.
- [x] **End-to-end coverage** — `internal/e2e` runs the real `httpapi.Server`, `library.Build` and `packcache` against the real client. Stubs prove one side's logic; only this catches the two sides disagreeing about the wire. All five tests were red-proofed by breaking the code they cover.

- [x] **Run it against the oracle** — done 2026-09-05. YARG v0.15.0 pointed at a folder `yarg-sync` filled: **20 of 22 accepted**, the two refusals being the two our scanner flags. "Unmodified YARG just sees files" is now a measured result rather than a claim. The run found the fifth oracle finding and two real defects — a `notes.mid` with no note tracks was not flagged at all, and `PreparseUltraStar` skipped note derivation whenever `#TITLE` was missing, so a chart reported zero parts for a song that plays. Both fixed and red-proofed; write-up in `docs/TEST-CORPUS.md`.
- [x] **Run the server as a container, off the development machine** — done 2026-09-05 on vault2. The published image had never actually been executed before this; CI only ever checked the image config's architecture and the binary's ELF machine type, which are checks on bytes. It now runs: 8.6 MB distroless, indexed 22 songs in 7 ms, `version=827e0fe`. `yarg-sync` from ENG-1 over Tailscale pulled 22/22 in 341 ms — recorded at the time as "byte-identical to the localhost run", which was a **byte-count** comparison and not a per-file one; the files were in fact not identical, for the reason below — and a second sync transferred nothing. **Unmodified YARG on the result: 20 accepted, 2 refused** — the same two, both flagged by our scanner. Write-up in `docs/DEPLOY-VAULT2.md`.

- [x] **Execute arm64 on real ARM hardware** — done 2026-09-06 on a **Raspberry Pi 4 Model B Rev 1.5**, `aarch64`, Debian 13 trixie. Both halves ran:

  - the cross-compiled `linux/arm64` **binary**, confirmed by the Pi's own `file` as `ELF 64-bit LSB executable, ARM aarch64, statically linked`, indexing 22 songs in 24 ms;
  - the published multi-arch **container image**, `arch=arm64`, image id `3c5fdd28…`, `version=1ae2219`, indexing 22 songs in 25 ms under docker.io 26.1.5.

  Until this run, "portable to Raspberry Pi" rested entirely on ELF machine type and image config — checks on bytes, never an execution. It is now a measured result.

  And it extended the determinism result. Archives served by the arm64 server, both as a bare binary and as the container, are byte-for-byte identical to all five earlier x86-64 sync sets. Seven independent syncs, two operating systems, two CPU architectures, 154 archives, one set of bytes. The mask derivation is architecture-independent, which it had to be and which nothing had tested. Write-up in `docs/TEST-CORPUS.md`.

  Caveats stated rather than buried: the board was a **Pi 4, not the 3B+** this project had been waiting on (that board is dead — steady red PWR, no ACT activity, with three independently written cards). The Pi was a **one-shot machine, wiped afterwards**, so this is "arm64 ran on this date", not a maintained deployment. And the image reached it by `docker save` on vault2 and `docker load` on the Pi rather than a direct registry pull, to avoid putting a registry credential on a throwaway host — the image id is identical either way, so what ran is the published image; only its transport differed.

  The one-shot shape is deliberate, not a shortcut. The Pi is a *verification target*, not a development or deployment one: development happens on Docker, and a Pi is picked up periodically to confirm the arm64 build still runs on real ARM silicon. So there is no maintained Pi to keep current, and no Pi-shaped work waiting on hardware — each arm64 claim is dated, and re-measured rather than inherited. - [x] **The exit criterion, with a second client** — done 2026-09-06. YARG was installed on r7-desktop by copying ENG-1's portable install, and both machines synced from one server and were scanned by unmodified YARG: **20 accepted, 2 refused on each, the same two songs for the same two reasons.** - [x] **Deterministic packing** — found by that run and only findable by it. The two machines received 22 archives each, 229,515 bytes each, 0 failures each — and **16 of the 22 files differed**. `sng.Write` drew its header mask from `crypto/rand` on every call, so `PackDir` was not a function of its input, and 16 was exactly the number the bounded cache had evicted and re-packed between the two syncs. That broke the strong `ETag`, broke a `Range` resume spanning an eviction, and made "two machines syncing one server get the same bytes" false

wherever this repo asserted it. The mask is now derived from the package hash (`sng.MaskKeyFor`); it is stored in plaintext in the header regardless, so deriving it gives nothing away. **22/22 identical across a full cache wipe and across the two machines.** Two of the things that hid it are worth more than the fix: a same-song-twice test that was always a cache *hit* and therefore passed whatever the packer did, and `TestWriteUsesAFreshMask` , which actively *required* the non-determinism. Write-up in `docs/TEST-CORPUS.md` . - [x] **Re-measure the containerised chain against the fix** — done 2026-09-06. vault2 re-pulled onto `c623dae` with its pack cache wiped, then five independent syncs compared file by file: ENG-1 and r7 against the working-tree server, ENG-1 against that server after a full cache wipe, and both machines against the container. **110 archives, all one set of bytes.** Two servers built for different operating systems by different toolchains agree, so **the mask derivation is not platform-dependent** — which it had to be, and which nothing had shown. Unmodified YARG on the container's output: 20 accepted, 2 refused, the same two. - [x] **Deploy the hardening and measure it on the real server** — done 2026-09-06. vault2 re-pulled onto `077f36e` , pack cache wiped, all 23 songs re-packed by the new binary: **byte-for-byte identical to the `423902b` cache**, and the 23 files a `yarg-sync` then received hashed to the same set as the 23 packs on the server. Every earlier determinism result compared machines or architectures at a *fixed* commit; this one holds **across a code change**, which is the case an `ETag` and a `Range` resume actually meet, since a server is upgraded far more often than it changes CPU.

And the defect was reproduced on the deployment before being declared fixed. A probe archive with backslash separators was dropped into the host library: the server indexed 23 songs with `problems=1` , naming the file and saying how to fix it, in the log and in `/api/v1/library` both. The previous image reported `problems=0` and said nothing. Probe removed, library restored to 23 clean cases. See `docs/DEPLOY-VAULT2.md` . - [x] **Measure what this costs at real library sizes** — done 2026-09-06, `cmd/mkscale` and `docs/SCALE.md` . Every number in this repository until now came from 23 songs and 238 KB. Now: **index is linear**, 0.594 / 0.583 / 0.569 ms per song at 1,000 / 10,000 / 31,109 songs, and memory fits **13 MB + 6.5 KB per song** across the whole range.

Those constants extrapolate, and one of them is a constraint on goal #1: **a 100,000-song library needs ~665 MB resident**, so a 1 GB Pi 3B+ could not hold it while serving from it and a 4 GB Pi could. The catalog is in memory by design (ADR-002); this is the number that says where that design runs out.

On the byte axis — 200 songs of 5 MB audio, 1.07 GB — **resident memory stayed at 16.5 MB while serving 1.17 GB**, so packs stream rather than being assembled in memory, which the code was written to do and nothing had measured. With the cache bound set to 100 MB against that 1.17 GB library, **all 200 songs were still delivered and the cache never exceeded 102 MB** — the `pack_cache_max` property re-measured under 11x pressure rather than trusted.

The library is generated, not downloaded: community charts carry copyrighted audio, and a corpus that cannot be rebuilt on another machine is not a corpus. **What this does not close:** the songs are uniform, so a real library's *variety* is untested, and the oracle has never run at scale — that is the measurement most likely to find a YARG rejection category the scanner misses, and it needs real charts. - [x] **The oracle, against REAL charts and as a script** — done 2026-09-07. `scripts/oracle.ps1` replaces a documented hand procedure: it repoints YARG's `SongFolders` , wipes the cache, launches, waits, compares both verdicts, and restores `settings.json` in a `finally` so a crash still puts the operator's YARG back. It waits on the **song cache** rather than `badsongs.txt` , because a library YARG is happy with writes no `badsongs.txt` at all, and it requires that file to be newer than the launch so last run's verdict can never be read as this one's.

Run against **128 real community songs** (1.16 GB, supplied by Jay): YARG refused 0, we flagged 0. **A weak positive, and recorded as one** — both sides agreeing means there were no disagreements available to find, and a curated pack people actually play is close to the least informative sample for this test. What it does establish is **zero false positives on real charts**, which nothing had tested, and that `ErrTooManySongs` is right about real pack `.zip`s rather than only synthetic ones. - [x] **Un-skip upstream's own scanner tests** — done 2026-09-07. `YARG.Core.UnitTests`' `FullScan` and `QuickScan` skip unless `YARG_TEST_SONG_DIRS` names a song directory. Pointed at the same 128 songs, both pass and YARG.Core writes no `badsongs.txt` — a second oracle that runs **in-process in seconds**, needing only the .NET SDK rather than the game. Upstream's own comment says the only fail condition is an unhandled exception, so it is a crash test over real input, not a verdict test; `docs/TEST-CORPUS.md` says so rather than letting it read as more.

Still uncovered: real-world `.chart` coverage is zero — and `.chart` is the format whose early-return bug this project already found once.

Re-measured 2026-09-07 and it is worse than "all 128 songs are `.mid`". The three multi-song `.zip` packs the scanner refuses were opened and counted too: **270 songs in the corpus, 128 of which are `.chart`-free loose folders and 128 inside the packs, and every single chart file is `notes.mid`. Zero `.chart`, zero `notes.txt`.**

Onyx is not the route, and the handoff that said it was should be ignored. Its release notes describe `.chart` as an *import* format and its conversions as producing CH-compatible output (`song.ini` + `notes.mid`) — the wrong direction. A converted chart would also be the converter's dialect rather than what charters actually publish, which is most of the point of wanting real ones.

What would actually close it: a handful of genuine community `.chart` songs dropped into `C:\dev\_incoming\YARG` — Clone Hero-era charts are commonly `.chart`. That is a two-minute job for a human and one this session cannot do for itself. - [x] **Break real songs on purpose, and find something** — done 2026-09-07. `cmd/mkbroken` damages songs with real structure in 16 named ways, each carrying a **predicted verdict**, so the oracle compares three columns rather than two: prediction, our scanner, YARG.

The standard failed, for the first time on this axis. Of 6 songs YARG refused, **2 were ones we passed with no issue at all**: a MIDI truncated to a third of its length, and a plain text file named `notes.mid`. Both draw "*Corruption of either the ini file or chart/mid file*" from YARG. The cause was written down that day as "*the scanner never parses the chart, it only hashes it*" — **and that diagnosis was wrong**; see the next entry, which is the more useful half of this one. - [x] **Decide what "is this even a chart?" should mean** — done 2026-09-07, and the answer was not the one that had been designed.

The proposal was a structural check: `MThd` magic on `notes.mid`, a `[Song]` section in a `.chart`. Probing eleven damaged shapes first — printing every issue rather than reasoning about them — showed that **half of it already existed and the other half would have missed the case that mattered.**

The scanner has preparsed every chart since Phase 1. For an unreadable chart it *detected* the failure and recorded a cosmetic `parts_note`, which is indistinguishable from not detecting it; that is now the issue `chart_unreadable`. For a `.chart` that is not a chart, `no_notes` already fired, so that half of the proposal was unnecessary. And the `MThd` check would have passed `03-chart-truncated` cleanly, exactly as the warning above said: a real nine-track chart cut to a third keeps a valid header and several **complete** `MTrk` chunks, so it preparses fine and reports genuine instruments.

The check that does work is the file contradicting **its own chunk lengths** — issue `chart_truncated`. It does not catch a cut landing exactly on a chunk boundary, which is byte-for-byte a valid chart with fewer tracks; that limit is pinned as a test rather than left as a comment.

Measured: 270 real songs, 2 flagged — precisely the two damaged on purpose. Oracle run 8: 6 refused, **0 missed**, standard held. - [x] **Measure what several clients at once do — and fix what it found** — done 2026-09-07. Every measurement before this was serial, while the whole point of the project is a server on a LAN with more than one client. The probe found a **real defect**: the song handler asked the pack cache for a *path* and opened it as a second step, so an evicting goroutine could remove the archive in between and the server answered **404 "this song is no longer where the index says it is; rescan"** for a song that was present. Measured at 64 clients over a 40-song library bounded to two archives: **2, 1 and 0 spurious 404s across three runs of 2,560 requests** — rare enough never to be met by hand, constant on a busy server, and it points the operator at a library that is fine.

`packcache.Open` now returns an **open file**, so there is never a moment where the caller holds a name but not a handle. Two properties held under the same pressure: the thundering herd collapses (64 concurrent requests for one cold song → one pack, one archive, 64 identical responses), and **eviction costs a re-pack and never data** — 7,680 requests against a cache bounded to a *single* archive returned byte-identical archives, zero mismatches. One promise was corrected: `pack_cache_max` is enforced after each insert, so it is a **high-water target rather than a hard cap**, overshooting by one to three archives as concurrency rises. Details in `ADR-002`; the instrument mistakes are in `SCALE.md`.

Resolved, and worth remembering for the response rather than the event: on 2026-09-05 a Defender machine-learning verdict (`Trojan:Win32/Bearfoos.A!ml`) quarantined `cmd/yarg-sync` builds for about a minute, then stopped reproducing within the hour with nothing changed on the machine — verified across three forced relinks and three distinct binary hashes, zero new detections, same signature version, no exclusion added. Acting on it immediately would have left a permanent Defender exclusion behind for a problem that had already evaporated. If it returns, re-measure before doing anything structural; the escalation is Microsoft's false-positive form, never an exclusion. Code signing is *not* an available escalation — it was considered and declined on 2026-09-08 (free software, no certificate subscription). Details in `docs/SYNC-CLIENT.md`.

Exit criterion: a Pi on the LAN serves a shared library; two machines running stock YARG both see the same songs without either one being modified.

Phase 3 — Native remote song source in the client fork

Only now does the `yarg/` fork get touched. This is the upstream-facing work.

- Add an HTTP song source alongside local folders — browse, stream, cache.
- Touches YARG's song cache, so it must be designed with upstream in mind rather than bolted on.
- PRs target `dev`. Never `master`.

Decided 2026-09-06: build it in the fork now, and approach upstream in parallel. Their answer shapes this work; it does not block it. That is a deliberate choice rather than impatience — their own contributing guide invites experimentation on undecided features and says large progress can promote a tier, so a working implementation is a better opening than a proposal.

Reconnaissance done 2026-09-06, recorded in `UPSTREAM.md` with the draft post:

- **Their `CONTRIBUTING.md` sorts every feature into six tiers, and the tier decides whether a PR is even read.** A remote song source matches none of the published examples, so *which tier* is the first question — and asking it on Discord before building is what their guide explicitly tells contributors to do.
- **Nothing upstream proposes this.** Searched their issues: the nearest is **#860**, a built-in web server for search and queueing from a phone — a *control plane*, not a content source, and worth not conflating with this. `#1030 "user-supplied song sources"` is about source **icons**.
- **Two of our three permanent non-goals are things upstream has independently ruled out.** CON Decryption is in their Out of Scope list, verbatim: *"Do NOT PR these features. Your PR will immediately be denied."* We are not asking them for anything they have already refused, and saying so is worth a sentence in the opening post.
- **The ask is smaller than it sounds.** The server already hands out plain `.sng` that unmodified YARG reads, so this is not "support our protocol to play our songs" — it is "fetch from a URL instead of needing a separate sync tool". The seam underneath is that `SongEntry` is abstract but `ActualLocation`, `SortBasedLocation` and `GetLastWriteTime()` all assume a local path; *a song entry whose bytes are not on disk* is a general capability rather than a feature about our server, and may be an easier thing for upstream to want.

A note, not a plan. The same client-to-server channel could carry a queue, which is the shape of upstream's #860. That is worth knowing when we eventually talk to their team, and it is not on this roadmap — see "What this project is, and what it is not" at the top. (An earlier revision of this paragraph called #860 unbuilt and said nothing could reach a running client. Both are false: #984 does exactly that, and has since February 2025.)

Design recorded 2026-09-07: `ADR-004`, grounded in YARG.Core rather than in the paragraph above. Reading the code changed the shape of the ask twice over, and the entry above was wrong in one place — it says the seam is that `ActualLocation` / `SortBasedLocation` / `GetLastWriteTime()` assume a local path. True, but not the useful finding.

- **YARG.Core contains no networking whatsoever** — `HttpClient|System.Net|UnityWebRequest` returns **zero** matches across the whole library. The Unity project does network (Discord, localization, song sources), but none of it produces a `SongEntry`. So this is not "extend the existing pattern"; there is no pattern, and putting an `HttpClient` in YARG.Core is a change of character rather than an addition.
- **`SngEntry` and `UnpackedIniEntry` are `internal sealed` with private constructors**, and so is their base `IniSubEntry`. A remote entry type cannot be added from outside the assembly at all — every entry-level approach is a change *to* YARG.Core.
- **Exactly one source seam already exists** — `protected abstract FixedArray<byte>? GetChartData(string filename)` — and it is not enough: `SngEntry` reaches for `SngFile.TryLoadFromFile(_location, …)` at seven sites and the local filesystem at nine more. `SngFile` has no stream-taking loader, though `FixedArray` does.

So the decision is **YARG.Core stays offline and the Unity layer fetches**, in three increments, with only the first committed to: a managed mirror folder inside the game (**zero YARG.Core changes** — it is `yarg-sync` moved in-process, and it needs no answer from upstream to ship); then lazy fetch-on-play, which needs exactly two named seams; then server-supplied entries, deferred because consuming that path means speaking the cache format, whose version constant is `26_09_04_00` and whose groups store absolute paths — the `songcache.bin` non-goal wearing a hat.

The upstream ask gets much smaller as a result: not "support remote libraries" but "take `SngFile.TryLoadFromStream`, and consider a materialise hook on `IniSubEntry`" — both small, both useful on their own, neither dragging networking into the library.

Increment 1 shipped 2026-09-07 — `yarg` `7590443` on `dev`. The game mirrors a server's library into `PathHelper.ServerLibraryPath` and appends it to the scan list beside `PathHelper.SetlistPath`; **zero YARG.Core changes**, as the ADR predicted. Verified against vault2 from batchmode, not merely compiled: 23 songs, 238,215 bytes, every chart hash checked with YARG.Core's own `SngFile` / `HashWrapper`, second run downloaded nothing. That byte count is identical to what four concurrent `yarg-sync` clients pulled from the same server, so the C# and Go clients agree exactly.

One thing the ADR did not predict, found by running it: **Unity's `insecureHttpOption` defaults to `NotAllowed` and a LAN song server is plain HTTP**, so the first smoke test failed outright. Set to `DevelopmentOnly` — enough to develop and test, nothing weaker shipped to players — and what a release build should do is now question 4 in the Discord post rather than a default quietly changed in a fork.

The same defect in `yarg-sync`, `63c4f31` — because the two clients mirror each other. Having found it in the in-game mirror, the obvious next question was whether the Go client did the same thing. It did: `postHave` returned the server's list straight to the caller and every entry became both a URL and a path. `filepath.Join` **cleans** a path but does not **confine** it.

The exposure differs by platform, and the difference is the interesting part. On POSIX the write is *transient* — `fetchOne` downloads to a `.part`, verification rejects the bytes, and `os.Remove` succeeds, so nothing is left to find. The identical sequence on Windows leaves the file, because YARG.Core holds a non-`.sng` file open and the delete fails. **Two bugs compose on one platform and not the other**, which is why the Go test asserts on the *request count* rather than on files left behind: a test whose real assertion never fires is decoration. Red-proofed by widening the guard — six song requests for names that are not chart hashes.

Go is better off than .NET in exactly one respect: `filepath.Join` does not discard its first argument when the second is rooted, so an absolute name cannot escape the way it does in C#.

`prune` was checked and is **not** affected — it iterates the local inventory, which the managed -name regex already filters, and only removes hashes the server does not have. A server cannot name a file for deletion.

Remote arbitrary file write in the mirror client, `yarg` `f851fd5f`. The most serious defect found in this project so far, and it came from following the hostile server one question further: the client asked the server what it was missing and used the answer as **filenames**, unchecked.

```
foreach (var entry in json.Value<JArray>("missing") ?? new JArray())
    missing.Add(entry.ToString());          // no validation
...
string part = Path.Combine(destination, hash + ".sng.part");
```

`../..` escapes the mirror folder, and an **absolute** path is worse, because `Path.Combine` discards its first argument when the second is rooted — the file goes exactly where the server said. **Demonstrated with the guard temporarily widened**: a 2,048-byte attacker-controlled file written to an attacker-chosen absolute path, persisting on disk. With the guard restored: `rejected=6`, nothing outside the mirror, sync still completes.

What was shown, precisely. Only the absolute-path case was observed to land, because that directory existed; the `../../../..` names resolved to directories that do not, and `DownloadHandlerFile` does not create them. That is a limit of the demonstration, not a defence — an attacker picks a directory that exists. The first version of the test proved nothing at all, because the server 404'd the hostile names and no bytes were ever served.

The persistence needs a second bug. The download succeeds so the file is written; verification rejects it; and the cleanup delete is the one that *cannot* succeed, because YARG.Core keeps a non-`.sng` file locked. Two known bugs compose into a durable write.

"The player chose this server" is not an answer. The connection is plain HTTP on a LAN by design, so anything on the path can supply that list, and a server trusted for *content* should still not be trusted to name files on every machine that syncs from it. The server-side scanner already refuses traversal entries inside archives for exactly this reason; **the client had never been given the same treatment.** That asymmetry is the lesson worth keeping — the guard was written once, on the side where the danger was obvious.

The per-song library badge, `yarg` 2026-09-08 — and the blocker turned out to be softer than it had been written down. Every handoff since increment 1 has said the badge "needs the Unity editor open, cannot be verified headless". That is true of a badge that is a new object on the song-row prefab. It is not true of the badge itself: `GetSecondaryText` already returns rich text (`FormatAs` wraps the artist in `<color>` and `<font-weight>`), so a tag appended after it needs no prefab change at all — and rich text CAN be measured from batchmode.

So the badge rides on the artist line. That is a stated limit rather than a first draft: the prettier version is a prefab change, and a prefab change can only be looked at, never asserted. This is the version of the feature that can be checked.

The decision lives in `SongViewType.WithServerBadge`, **static and pure on purpose** — a `SongViewType` needs a `MusicLibraryMenu`, so an instance method would have been untestable headless, which is precisely how the feature came to be labelled unverifiable in the first place.

`Editor/LibraryBadgeProbe.cs` plants two corpus songs in the game's REAL mirror path (`PathHelper.ServerLibraryPath`) and two in an ordinary folder, scans both with YARG.Core's own `CacheHandler.RunScan`, and asserts the badge lands on exactly the mirrored ones. Entries built by the real scanner, not by hand: a hand-made entry would only prove the string comparison works.

Red-proofed in both directions, because a marker that appears on everything and one that appears on nothing fail differently:

Change	Result
badge every song	"*2 of 2 songs the player owns were marked as the server's*", and the null-entry assertion caught it too
badge no song	"*2 of 2 mirrored songs are unmarked*" alone

The first version of the probe failed for a reason that had nothing to do with the badge, and it is the harness trap this project has already paid for once. It took the first four `.sng` files in the corpus; two of them vanished from the scan and the probe reported "*1 mirrored and 0 local entries*". `badsongs.txt` named the real cause: "*Corruption of either the ini file or chart/mid file*" — the message YARG produces when `song.ini` omits `song_length`, the scanner falls back to measuring the audio, and **batchmode has no working audio backend**. The probe now picks its songs by scanning the corpus first and using what survives, and exits **2 — inconclusive** rather than red if too few do.

Measured while doing it, and worth knowing before designing any other batchmode test: **only 5 of the 23 corpus songs scan in batchmode at all.** The other 18 are refused for reasons that are the harness's, not theirs.

The badge missed the one song that is most obviously the server's — 2026-11-02

Both the badge and the `server:` filter decided via `MirroredSongs.IsMirrored`, which asks whether the song's bytes sit under the mirror folder. Both shipped **before** `RemoteSongEntry` existed, and *Browse and play server songs without mirroring the library* (`yarg` `36ca1f2a`) touched neither file — so a song the player can see and has not downloaded was the one song treated as local. Its `ActualLocation` is the synthetic `server://<chart hash>`, which is under no folder at all.

Measured against a real 6-song catalogue from vault2, entries built by `RemoteCatalog.FromJson` — production's own mapping, because a hand-made entry would only prove a string comparison works:

	Before	After
browsed songs carrying the badge	**0 of 6**	6 of 6
`server:yes` keeps them	**0 of 6**	6 of 6
`server:no` excludes them	not asserted	6 of 6
mirrored songs badged / local songs clean	2 of 2 / 2 of 2	unchanged

The filter half is the worse of the two. A missing badge is cosmetic; a player who types `server:yes` to see what the server has, and is shown only what they already downloaded, has been answered the opposite of what they asked.

`MirroredSongs.IsFromServer` is now what both call — `entry is RemoteSongEntry || IsMirrored(entry)`. `IsMirrored` keeps its narrower meaning, because the settings row that counts what this machine holds genuinely wants "bytes on this disk".

The type test is not a retreat from "the signal is the path". For a song with no bytes here there is no path to read, so the entry's type is the whole of the available evidence — and it is conclusive, because a `RemoteSongEntry` exists only because the server published it. Once such a song is fetched the path test answers too, and the two agree. Recorded as the one judgement call in the change, so it can be argued with rather than discovered.

Two launcher quirks that are exact opposites, and one probe that skipped instead of failing

The new assertion's first run reported `PROBE SKIP: no catalogue at YARG_CATALOG_JSON, so browsed-song behaviour was NOT measured`. `Invoke-CimMethod Win32_Process.Create` spawns through the WMI service rather than forking the calling shell, so **it does not inherit that shell's environment** and the `$env:` set beside it never arrived. The fix is a `.cmd` wrapper that sets the variable and launches Unity as its own child.

That is worth writing down twice over. First because the assertion did not fail — it did not *run*, and only reported so because whoever wrote the skip path made it say `NOT measured` out loud instead of passing quietly. A default-to-pass would have had the defect written up as absent.

Second because it is the **opposite** of the launcher lesson from the same day's oracle run, where `Win32_Process.Create` produced a 97-byte `songcache.bin` and `Start-Process` was correct (`docs/TEST-CORPUS.md`, eleventh run). Unity needs `Create` because a bridge-spawned process inherits a job object that kills its Package Manager child; YARG.exe

needs `Start-Process`. Same two functions, opposite right answers, unrelated reasons — so neither generalises, and "use the launcher this document names for THIS program" is the only safe rule.

And a third, smaller: in PowerShell `@('a', 'b' + $x + '"')` does **not** build two strings. The comma binds tighter than `+`, so it builds `@('a','b')` and then appends `$x` and `"` as further elements. The generated `.cmd` had `set "VAR=`, the path, and a lone quote on three separate lines, and Unity never launched. Parenthesise each element.

The last unchecked server string: `package_hash`, both clients, 2026-09-08. Closing the file-write defect above left one server-supplied string still used without looking at it. When a chart hash exists in two packages the server answers **300** with the candidates, and the client puts the one it picks straight into `?package=` on the request that follows — unchecked, in C# and in Go alike.

Severity is genuinely low, and saying otherwise would be inflating it. The value reaches a URL, not a filename, and `VerifyChartHash` still rejects whatever comes back if it is not the song that was asked for. What it could do is put `&`, `#` or whitespace into a request the client makes. It is closed because it was the last one, not because it was dangerous — the interesting property is that *the same class of bug was written three times, in two languages, by trusting a server for names as well as for content*.

Both clients now check `^[0-9a-f]{16,128}$` — the same pattern `packcache` already enforces server-side — and both **skip** a malformed entry rather than failing the song, which matters more than it looks: skipping keeps the two clients choosing the *same* package, which is the entire reason the choice is deterministic. A 300 with nothing usable fails, and says so.

Red-proofed on both sides, because a check nobody has seen fail is a check of unknown meaning:

Side	Check disabled	Result
Go	`if false && !packageHash.MatchString(...)`	`TestPackageHashesThatAreNotHashesAreRefused` alone failed: *client chose [../../../../etc/passwd]*
C#	`PackageHash.IsMatch` removed from the guard	`HostileServerProbe` failed both new assertions and nothing else: the client asked the server for `../../../../yarg-probe-escape/package`, and for the all-bad listing it asked for the **empty string** — an entry that sorts below everything, which is the sort of value a check written by reasoning alone tends to miss.

The probe's hostile 300 lists `../../../../yarg-probe-escape/package` **first in sort order** (`.` is 0x2E, below `0` at 0x30), so a client that skipped the check would pick it: the assertion fails loudly rather than by luck. A second 300 lists nothing usable at all, and the client must fail that song without inventing a request — `RequestedPackages` on the fake server records what was actually asked for, rather than the probe assuming.

One existing test and the probe both had to be **corrected, not just extended**: they used toy package hashes (`aaaa`, `0001`) that the new check rightly refuses. Toy values were testing a path no real server can reach.

The sweep was broader than the guarantee, `yarg` 2026-09-08. Auditing what the mirror DELETES, rather than only what it writes, turned up a smaller mismatch in the same family. The in-game mirror never prunes and `yarg-sync`'s prune iterates the LOCAL inventory, so **a server cannot name a file for deletion in either client** — that part held, and

is now stated in `SYNC-CLIENT.md` rather than left to be re-derived. But the mirror's sweep of dead partial downloads took **any** name ending `.part`, while the guarantee three lines above it says anything not named like ours belongs to the player and is never touched.

The mirror folder is one the game owns, so a stranger's `.part` in it is unlikely; the claim was still wrong, and **a guarantee that holds "almost always" is not one**. The sweep now matches `^[0-9a-f]{40}\.sng\.part$`, exactly what the download path can write, and anything else counts as the player's. The probe plants `stranger.part`, `not-a-hash.sng.part` and one ending `.sng.part.bak`, and asserts all three survive a sync against a server that keeps failing.

And the instrument was wrong before the code was, for the sixth time. Planting those files made an existing assertion fail — "*3 dead .part file(s) were there to sweep but only 1 were swept*" — which reads exactly like a broken sweep. It was `Directory.GetFiles(destination, "*.part")` counting the player's files as ours. The measurement was fixed, not the code.

The browse page was audited in the same pass and is a non-finding. It renders `song.ini` metadata from uploaded archives — content the server does not author — and `card()` escapes every field, with `encodeURIComponent` on the download link. Exactly three interpolations bypass `esc()`: a `problems.length` count, a `URLSearchParams.toString()`, and `p.intensity`, which is a Go `int8` and so cannot marshal as a string. Recorded in `API.md` as a negative so it is not re-investigated, with a note that the third is the one to re-check if `intensity` ever stops being an integer.

The download error branch was dead code, `yarg` `48bcb598`. Following the hostile probe with one more question — what does a *player* read when a download fails — turned up that they read `Unknown Error`, and then that the branch meant to say more had never run.

UnityWebRequest reports a non-200 in two ways and the difference is not obvious. A 4xx/5xx or a dropped connection makes UniTask **throw** from the await, so everything written after it — the `if (request.result != Success)` check this code used to report failures — is **unreachable**. A **300 comes back as success** with `responseCode 300`, because there is no `Location` header to follow, so the duplicate-package case must be checked after a normal return. Both are now handled on both paths so neither assumption is load-bearing. A cut-off download now says *"the connection ended after 4317 of 8634 bytes (HTTP 200)*".

The 300 path had never met a 300. The live corpus has `duplicate_packages: 0`, so nothing had ever served this client one. The probe now does — declining to choose, listing two packages, and serving bytes only when asked for one by name — and it earned itself immediately: the first version of the fix assumed a 300 also arrived as an exception, which broke the duplicate-package path. **A regression introduced and caught inside the same change**, in a path that would otherwise have made every song existing in two packages permanently unfetchable, silently, until somebody had a duplicate.

The mirror's integrity guarantee, tested against a hostile server. `SongServerSync` claimed in its own comment that "a crash or a dropped link mid-download cannot leave a truncated archive under a name the scanner will trust" — true by inspection, never reproduced, which is precisely the standing the packcache eviction race had until CI ran the one test that could fail. `Editor/HostileServerProbe.cs` now serves the four ways a download goes wrong from a raw-socket server: a body cut in half mid-transfer (Content-Length promises the whole file), a real archive served under someone else's hash, a 500, and random bytes.

The guarantee held — no bad archive was ever named — but three real defects came out around it, none of which any unit test had reached:

1. **The cleanup delete could throw and replace the real error.** A rejected download reported "the process cannot access the file" instead of why it was rejected. The delete is now guarded and can never mask the cause.
2. **`SngFile.TryLoadFromFile` leaks its `FileStream`** when the file is not a `.sng`, so on Windows the rejected file stays locked and cannot be deleted at all. Partial downloads are now swept by `Inventory` on the next run instead, and `Result` reports `swept=`.
3. **`SngFile.Dispose()` throws on the value a failed load returns**, and that `NullReferenceException` replaced our own message — "downloaded file is not a readable .sng" reached the caller as "Object reference not set to an instance of an object".

Two of the three are upstream's, both one-liners in the same failure path, and both are now written up with reproductions in `UPSTREAM.md` as something to hand them ahead of any API request.

The probe also pins what the guarantee does **not** promise: a failed download can still leave a `.part`, because the locked file cannot be deleted. What must hold is that a `.part` is never a song and that dead ones do not accumulate — measured at 1 after one sync and 1 after two, against a server that keeps failing.

A `server:` search filter came first, `yarg` `b2629b6e`. `server:yes` shows only what the mirror brought in, `server:no` only what was already yours, and it composes with everything else (`artist:queen;server:yes`).

The signal is the path, and that was measured before anything was written. A probe scanned a real 23-song mirror and printed what a mirrored song looks like to the library. Both obvious candidates fail: `Source` is "Unknown Source" — it belongs to whoever charted the song — and `Playlist` is "Unknown Playlist" for every one, because a loose `.sng` takes its playlist from its own metadata rather than the folder it sits in. Playlist matched the mirror folder's name for **0 of 5** scanned entries; `ActualLocation` was under the mirror for **5 of 5**.

That also answered the question worth asking first: **YARG already has a `folder:` filter, and it matches on playlist** — so `folder:serverlibrary` does not do this today and the new filter is not a duplicate. Had it come back the other way, the right move would have been to write nothing and document the filter that already existed; this project has already once designed a fix whose second half turned out to be unnecessary.

Not a new `SortAttribute`, deliberately. The search pipeline keys filters on that enum, but the enum is also what the library sorts and groups by and several values already have no comparer; adding one for something that can never *be* a sort order would put a value in it that half the code must remember to ignore. `server:` is lifted out of the query before the pipeline runs and applied as one pass over the already-narrowed result.

The per-song badge shipped after this was written, `yarg` `5865c24f` — and the paragraph that used to stand here said it could not be verified headless, which stayed on the page as an open item for three days after it was done. Pulling the decision into a static (`SongViewType.WithServerBadge`) is what made it measurable; see below. The claim "this needs UI work that cannot be checked" was true of the *prettier* version only, and it was load-bearing enough to be worth re-testing rather than inheriting.

A Song Server tab came first, `yarg` `011d688f`. The feature now has its own settings tab rather than three rows wedged under a header on Song Manager: URL, reachability, what this machine holds, the startup toggle, and Sync / Cancel. During a sync the mirror line becomes live progress.

A tab rather than a Main Menu screen, decided on cost. Menus are `MenuObject` children inside `Scenes/MenuScene.unity` keyed by a `MenuManager.Menu` enum, so a Main Menu entry means hand-building a GameObject subtree in a scene file — the one kind of change in this work that cannot be checked by measuring, only

by looking at it. A tab follows the `SongManagerTab` precedent and inherits navigation, layout and controller input. **It needs no new prefab**: every row is a type that already existed, including the live text row added with the status line and a `ButtonRowMetadata` that already took `params string[]`.

Two new kinds of check came out of it, both worth reusing:

- **Tab icons are validated against the real sprite atlas.** They are Addressables lookups by string (`TabIcons[Import]`), so a name that is not in the atlas compiles, runs, and produces a tab with no icon that nobody notices until a screenshot.
- **The mirror row is counted against a folder that really holds songs**, not only the empty case: "Mirrored: 23 songs, 233 KB on disk" against 23 files, and 238,215 bytes is 233 KB.

End-to-end against vault2 after the `Sync` signature changes: `server=23 had=0 downloaded=23 failed=0 bytes=238215` — byte-identical to the baseline from when the mirror first shipped.

Server status in the menu came first, `yarg` `69d7d007`**.** Settings -> Song Manager shows a live line under the URL: connected and how many songs, or not reachable and why, plus the last sync's outcome. Before this, the only way to learn whether a URL worked was to press Sync and read an error dialog — a typo, a sleeping NAS and a healthy server were indistinguishable until you committed to a sync.

It reports **the server's own health too**, because `/api/v1/library` already names every directory the scan could not read and a library that quietly indexes 9,000 of 10,000 songs looks exactly like a library that has 9,000 songs. It shows `distinct_charts`, not `songs`: the server counts packages, two packages sharing a chart are one song to YARG, and the package count would promise more songs than a sync can deliver. Verified against vault2 — the row reads "connected, 23 songs" and the server independently reports `distinct_charts: 23`.

One probe in this change went green while testing nothing, and that is the reusable part. `SettingsManager.Settings` is null outside a running game; the status check dereferenced it before setting any state, and the `NullReferenceException` was swallowed by a fire-and-forget call. The state never changed, so both assertions held vacuously — the log carried four NREs while the probe reported PASS. The fix was not only to null-guard the read (a real defect: the row is reachable from a menu drawn before settings load, and would have stayed blank forever while looking like it worked) but to make the probe call the same path directly, with nothing able to swallow an error. **A probe that cannot fail is not evidence**, and the tell was in the log the whole time — lesson 5, "a green must come from the thing being tested", found again in a new disguise.

Auto-sync on startup came first, `yarg` `7996c46b`**.** Pointing the game at a server is now something you do once: `Sync On Startup` (default on) mirrors before the startup scan, so a machine that plays does not have to be administered. Default-on costs nothing until a URL is set - the path returns before opening a socket.

Reading YARG.Core is what made it work, and the fact is worth keeping: a startup **quick scan cannot see new files.** `CacheHandler.QuickScan` only deserialises `songcache.bin` - its own summary is "*performing very few validation checks ... for the sole purpose of speeding through to gameplay*" - and never walks the filesystem; it falls through to a full scan only when it parses **zero** entries. Syncing before the ordinary startup scan would therefore have downloaded songs that stayed invisible until the player manually refreshed, and the feature would have looked broken while working perfectly. The scan mode is now decided by what the sync did: full when something arrived, quick otherwise.

Startup is the one place this must be neither loud nor slow. A song server is somebody's Pi or NAS and will be off, asleep or behind a dropped link a good fraction of the time, which is the ordinary case rather than the exceptional one. Every startup failure is logged and swallowed - no dialog in front of a loading screen - and reachability gets its own budget separate from download time: `Sync` takes `listTimeoutSeconds`, and startup passes 5 instead of 30. **Measured at 5.1 s** by `Assets/Editor/StartupReachabilityProbe.cs` against `192.0.2.1` (TEST-NET-1, RFC 5737, guaranteed unroutable). Deliberately not a closed port on localhost: a refused connection returns instantly and would pass the test while proving nothing, where an unroutable address hangs, which is what an unplugged Pi actually does.

The settings row came first, on 2026-09-07. Increment 1 shipped with `SongServerUrl` as a hidden field edited by hand in `settings.json`, because the project had exactly one `AbstractSetting<string>` — the IPv4 one — and its visual parsed IPv4 addresses itself. It is now a real row on the Song Manager tab:

- `TextSetting` is the shared base; the setting owns what is valid, the visual owns the text field and knows nothing about either.
- `IPv4SettingVisual` became `TextSettingVisual`, **renamed rather than replaced so its `.cs.meta` GUID survives** — that GUID is the prefab's reference to it and to its four serialized fields, and re-creating the file would have silently emptied all of them.
- Every text setting shares the one prefab. No Addressables change, no cloned asset.

The reuse plan was wrong in one place, and only reading the asset found it. The shared prefab has `m_CharacterValidation: 6` (Regex) with `m_RegexValue: '[\d.]'` baked in, so a URL is not merely rejected in that field — it is **untypeable**, the letters silently swallowed. TMP exposes no setter for `m_RegexValue`, so the filter is now installed through the public `onValidateInput` delegate, which takes precedence over the serialized one (`TMP_InputField.cs:631`, `onValidateInput ?? Validate`).

Verified by `Assets/Editor/SettingsRowProbe.cs` in batchmode, 9 checks, zero `error CS`: the prefab resolves `TextSettingVisual`, all four serialized references survived the rename, `onEndEdit` still reaches `OnTextFieldChange`, the localization keys are in `Settings.Setting` and not a neighbouring section, **a URL is typeable in the row**, and the IPv4 row still rejects letters and accepts digits.

One incidental measurement, upstream's behaviour and not ours: `IPAddress.TryParse` reads a leading-zero octet as **octal**, so typing `010.1.1.1` into the RB3E or sACN row silently stores `8.1.1.1`. The probe pins it so a future change to it is visible. A comment in `IPv4Setting` claimed the opposite until this was measured.

Exit criterion: the fork can browse and play from a server without a sync step, and a discussion thread exists upstream.

Half met, and the built half is finished rather than merely working. The game mirrors from a server with no separate tool, syncs on startup by default so the step is invisible rather than absent, and the whole feature is administered from its own settings tab — URL, reachability, what this machine holds, progress and cancel — with `server:yes` filtering the library to what the mirror brought in. The integrity guarantee is tested against a server that lies rather than asserted in a comment.

The discussion thread is still waiting on Jay to post, and what it asks has changed: a working implementation plus two `git am`-ready bug fixes, rather than a proposal.

Phase 4 — Modular features and the server config menu

The point at which the server stops being one feature and becomes a platform.

- ~~Feature registry: each capability is opt-in and independently enable-able.~~ **Built 2026-09-08.** `config.Resolved` now carries, alongside the settings, **where each one came from** — `default`, `file` or `flag` — and `Features()` turns that into one ordered list of capabilities. It feeds two consumers so they cannot disagree: the startup log, which now reports **every** feature including the ones that are off, and `GET /api/v1/features`.

Two things this fixes that were not on anyone's list. **The server never named the config file it read**, so an operator who edited a file the server never opened — wrong path, wrong working directory, a bind mount that landed elsewhere — got a server behaving exactly as if the file did not exist and saying nothing about it. And **a feature that was OFF logged nothing at all**, so "off because you turned it off" and "off because nothing you wrote was ever read" were the same silence. Both are the shape this project keeps paying for: *detecting something and reporting nothing is indistinguishable from not detecting it.*

Provenance is recorded by the parser rather than derived by diffing the config before and after, because a file setting a key to the value it already had is **still the file speaking** — and that is precisely the case an operator asks about. A test pins it.

The registry is **descriptive, not the gate**: the route is still registered from the resolved config, and a test asserts the registry and the routes agree in both directions, because a page reading "check_uploads: enabled" and then getting a 404 turns a configuration mistake into an apparent server bug. Red-proofed: breaking `Features()` fails tests in all three packages.

Paths and the listen address are deliberately **absent** from the registry. They are settings, not capabilities, and the endpoint is unauthenticated — which capabilities exist is already discoverable, a server's filesystem layout is not. Asserted, because "just return the config" is the obvious future shortcut. - ~~Config UI in the server app, so a user turns on only what they want.~~ **Built 2026-09-08.** `PUT /api/v1/features/{name}` changes a capability with no restart, and the browse page carries a settings panel with a switch per feature. That closes the last clause of goal 1.

`config_writes = off | local | lan`, default `off`. Not a bool, and that is the decision worth keeping: "yes" would have had to mean `lan`, and on a home network that is every device including the ones nobody is thinking about. This server has no authentication, so `lan` means exactly what it says and belongs to an operator who chose it on purpose. `local` — a settings page that works at the machine and not from a phone — is the useful middle.

Three properties, each asserted rather than assumed:

- **A disabled feature is still ABSENT, not forbidden.** That used to be true by construction: the route was never registered. A runtime toggle cannot work that way, because a route that does not exist cannot be switched on, so the routes are always registered now and the handlers answer with `http.NotFound`. The observable contract is unchanged and only the mechanism moved. The test compares a disabled feature's WHOLE response — status, body, content-type — against a path that genuinely does not exist. A status-only check would have passed against a JSON error body, which would still have told the caller the feature was there.
- **The write surface cannot widen itself.** `config_writes` is not writable. One call turning `local` into `lan` would end the "only from this machine" promise, made by whoever was already inside it.

- **Locality comes from `RemoteAddr`, never `X-Forwarded-For`.** A header the caller supplies would make `local` a suggestion: anybody on the LAN types one line and is treated as sitting at the machine. Red-proofed — trusting the header fails the test.

A change is written back to the config file, so it survives a restart — added in a second increment, deliberately separate because editing an operator's commented file in place carries its own risk. `config.SetKey` changes one line and leaves every other byte alone.

The rule that took thinking: **last one wins in this format**, because `Apply` walks the file in order, so the LAST uncommented line for the key is the one rewritten. Rewriting the first would leave a later line silently overriding it, and the operator's file would say one thing while the server did another. Red-proofing "first match" fails exactly that test and nothing else, which is what makes the test worth having. With no uncommented line the setting is appended with a dated comment, leaving the commented template — which is documentation — untouched. The value is parsed back before anything is written, because persisting a line that stops the server starting next time is the worst outcome a convenience feature can have. Atomic via a temp file in the same directory; permissions and line endings preserved.

A failed save is not a failed toggle. The runtime change stands either way and the response reports the two halves separately. The server never *creates* a config file: settings written into whatever directory it happens to be running from would land somewhere nobody would look.

A test caught a bug that reading the code twice had not — the rewrite emitted the captured pre-`=` whitespace *and* a space of its own, giving `check_uploads = yes`. The alignment there is the operator's, not ours.

Verified through a restart, which is the only thing that proves persistence: clicking the page's own switch wrote `check_uploads = yes` under a dated comment, the template's `# check_uploads = no` survived with all 73 comment lines intact, and the restarted server reported `feature=check_uploads enabled source=file`. That last word is the confirmation — the registry's provenance says the value came from the file, because it now does.

The page **asks whether it may write** rather than assuming, the same rule as the drop zone, and re-reads the server after a change rather than trusting what it just sent. Verified in a real browser both ways: with `local`, clicking the page's own switch flipped the server and the drop zone appeared with no reload; at the default, the panel showed state, offered **zero** switches, and a PUT from that page answered 404. - Candidate modules: multi-user libraries and permissions, playlists/setlists shared across clients, scores and leaderboards, library health reporting.

The web UI: managing the library without launching YARG

This is goal 2 from the top of the document, and it is the half of the server most easily mistaken for a nice-to-have. A library that can only be inspected through the game is a library you can only inspect on a machine with the game installed, sitting in front of the TV, with the game running. Everything else — checking whether last night's ingest worked, finding out why four songs are missing, seeing what a charter actually named a track — means launching a rhythm game to read a list.

Shipped 2026-09-07. `GET /` serves a phone-friendly page: search, the twelve sort attributes read from `/api/v1/library` so the page cannot drift from the server, and a per-song panel with metadata, parts, the scanner's issues and a download link. Embedded in the binary — no CDN, no build step — because a Pi at a party has no internet to fetch a stylesheet from. On by default (`browse_ui`); it exposes nothing the API did not already.

Registered as `GET /{$}`, and that detail is load-bearing: a bare `GET /` in Go's ServeMux is a catch-all that would answer every unmatched path with the page and a 200, turning every documented 404 into HTML a sync client would try to parse as a `.sng`. Red-proofed.

It rests on `/api/v1/songs`, which already does free text across name, artist, album, genre, subgenre, charter, source and playlist, with twelve sort attributes, ordering and paging, and answers a 10,000-song catalog in ~140 ms.

Where this goes next, all of it server-side and none of it needing the game:

- ~~**Library health as a view, not a log line.**~~ **Done.** The page fetched `/api/v1/library` for its sort attributes and threw `problems` away, so a library missing a thousand songs looked exactly like a library that had a thousand fewer. It now shows "N items could not be read", collapsed by default with each path and reason inside — a healthy library must not be made to look alarming, and the count is the part that matters at a glance. Paths and errors come from the filesystem, so both are escaped. The page is now `browse.html`, not `party.html`: the name was left over from framing this project does not own.
- **Ingest from the browser** — drop an archive in, watch it scan, see the verdict. **Half of this shipped 2026-09-08: `POST /api/v1/check` answers the verdict and keeps nothing.** It exists because the question people actually arrive with is *"why won't this song work?"*, and until now the only way to ask it was to copy the file onto the server, rescan, and read the log — a shell session, on the machine, for something the scanner answers in milliseconds.

Off by default, unlike `browse_ui`, and the difference is the decision worth keeping: the browse page is a new *view* of surface the API already served to anyone who could reach the port, while this accepts a large upload and spends CPU and temp disk on whoever asks. An existing deployment must not acquire that by being upgraded. When it is off the route is not registered at all, so the answer is 404 — a 403 would tell the caller the feature is there.

The verdict comes from `scan.ScanFile`, extracted from `WalkLibrary` in the same change so that the dispatch deciding *what a file is* has exactly one implementation. Two would agree for a while and then quietly stop, and the disagreement would read as "the checker said it was fine and the library dropped it" — the failure this project has already paid for in the packcache path and in both sync clients. A test uploads a real `.sng` and compares its chart hash against a library built from the same bytes.

The traversal test had to be written twice, and the first version was green for the wrong reason. It listed the staging directory's parent before and after — and passed against an implementation that really did join the uploaded name onto the staging path, because that implementation created the file outside and then the handler's own cleanup deleted it again before the test looked. Identical listings, green test, a server writing exactly where it was told. The damage `os.Create` actually does is TRUNCATION of a file already there, so the test now plants a sentinel where a traversal would land and asserts it still says what it said. Both plausible buggy implementations fail it now, and the failure message shows the operator's file being deleted outright. Same shape as the first hostile-server test, which proved nothing because the server 404'd the names it was meant to serve — **the second time in this project that a security test passed against the vulnerable code.**

The drop zone followed the same day. An endpoint with no UI still means a `curl` command, which is not "managing the library without launching YARG" for anyone who is not already at a shell. `GET /` now offers a drop zone, and `/api/v1/library` gained `check_uploads` so the page can ask whether the server will answer rather than assume it — capability from the server, the same rule that already governs `sort_attributes`. Files go one at a time: a dropped folder can be hundreds, and the target is a Pi.

Driven in a real browser against a real server, because reasoning about escaping is exactly how an audit goes wrong. Files were pushed through the page's own input handler, not through `fetch` written for the occasion: a real `.sng` pulled out of that library came back accepted with a matching hash and "Already in this library"; `garbage.sng`, `holiday-photos.rar` and `Some Song_rb3con` were each refused with their own reason; a file named `<img src=x onerror="document.title='PWNED'">.sng` injected **0 elements** and left the title alone; and the staging directory was **empty** afterwards, so "keeps nothing" is now measured on a live server rather than in a unit test.

Upstream contact is deferred to the bottom of the queue (Jay, 2026-09-08): nothing goes to YARG's team until there is something to show, and that means a public beta. See `UPSTREAM.md`.

And upstream is a collaborator we would welcome, not a gate we are waiting at — Jay, the same day. ADR-004 increments 2 and 3 are ours to decide, and the first thing that produced was a measurement rather than a decision: **increment 2 does not need `YARG.Core` after all, and should be SKIPPED anyway.** An entry survives a quick scan with its file deleted, so the Unity layer could fetch on play with no upstream change — but every player-facing refresh in the game is a FULL rescan (four call sites, all `quick: false`), which drops every unfetched entry, and the game's own "*Chart requires a rescan!*" path sends players straight at the button that does it. Increment 2 is increment 1 plus a trick the game itself undoes. The `YARG.Core` conversation belongs at increment 3, where entries would come from the scanner rather than from the cache.

Storing an upload is ADR-005 increment 2 and is deliberately not built: `--songs` is mounted `ro` on the live deployment, an unauthenticated write endpoint is a different risk from an unauthenticated read one, and the in-memory index (ADR-002) would need a rescan path. None of that is hard; it is just undecided, and increment 1 is useful without any of it. - **Metadata and playlists**, once multi-user libraries exist (Phase 4 proper).

What it deliberately does not do: control a running game. No queue, no remote select, and no UI hinting at either. That capability lives inside YARG and upstream has both an open request (#860) and a stalled draft PR (#984) in that space — see `UPSTREAM.md`, where it belongs as collaboration context. If we ever want it, it is a conversation with their team, not a feature we bolt on from outside.

Shipping the client — measured 2026-09-08

Nobody had ever built this fork. Every measurement until now was against the official v0.15.0 release or in editor batchmode, and "it compiles headless" is not "it ships". A public beta means somebody who is not Jay running the client, so the first question is whether a build exists at all.

It does. **526 MB in 34.1 minutes**, Unity 6000.3.5f2, `StandaloneWindows64`, Mono backend, and the fork's own code verified present in `Assembly-CSharp.dll` rather than assumed — `SongServerSync`, `MirroredSongs`, `SongServerTab`, `WithServerBadge`.

Three things came out of it that matter beyond "it worked", all in `docs/BUILD.md`:

- **Upstream has no Windows build workflow at all**, and its one mac workflow pins Unity `2021.3.21f1` against the project's real `6000.3.5f2`. There was no recipe to copy.
- **Unity returned `Success` WITH three errors.** All `RenderTexture.Create failed`, from `-nographics`. Almost certainly harmless, and "almost certainly" is the honest strength of that claim, because nobody has launched the build. The reporting now says `PASS WITH 3 ERROR(S)` rather than calling it clean — `BuildPipeline.BuildPlayer`

returns `Succeeded` alongside a non-zero error count, so a script that only read the result would have lied.

- **The binary identified itself as `YARC` / `YARG`**, upstream's own values — so a distributed executable said nothing about being a fork, and upstream would have fielded our bug reports. **Settled 2026-09-08** (fork commit `eff5a05b`): `FatalException` / `YARG-FE`. The label was the smaller half; Unity derives `Application.persistentDataPath` from those strings, so the fork had been writing `settings.json` and `songcache.bin` into an official install's folder. `Assets/Editor/IdentityProbe.cs` asserts the *moved folder*, not the edited strings, and also asserts that `PathHelper.LauncherPath` did NOT move — that one points at the external YARC Launcher and must keep doing so. See `docs/BUILD.md`.

Shipping the server — measured 2026-09-08

There was no Windows app, and there had never been one. Measured rather than assumed: no tray, service, GUI or installer code anywhere in the repo, and no GUI dependency in `go.mod`. What a Windows user could get was a console `.exe` inside a CI artifact that **expired after a week**, run from a terminal with flags. That is a build output, not something to hand anyone.

`cmd/mkrelease` now produces the download. One archive per platform — `.zip` for Windows, `.tar.gz` elsewhere — each carrying **both binaries**, a per-platform README, the config template, the licence and a launcher, inside a folder named after itself. Plus `SHA256SUMS`. Artifacts from `main` keep 30 days; a tagged build keeps its archives permanently.

Decisions worth keeping:

- **Go, not `zip` and `tar`.** The runner is a shell executor and neither tool is guaranteed to be on the host. "The packaging step failed because the runner lacks a binary" is worth designing out rather than discovering.
- **The shipped `yarg-song-server.conf` IS `config.Example`**, not a copy kept beside it. A second copy of a settings file agrees with the real one for a while and then quietly stops, and the person it misleads is the one who trusted the file in the download.
- **The packaging is deterministic** — every entry is stamped with a fixed date rather than the clock, so the same binaries always produce the same archive. Otherwise `SHA256SUMS` says nothing: every rebuild would differ, and a real change would be indistinguishable from one.

That is a claim about the packaging, not the whole build, and chasing the difference found two real defects — neither of them the one first blamed.

The local and CI archives for `4ddd322` did not match. First hypothesis: **CI pinned Go 1.27.1 and ENG-1 had 1.27.0**, a drift the CI comment ("keep this in step with the toolchain the workstations use") exists to prevent. Real, and fixed — ENG-1 is now on 1.27.1, installed the same verified way (official zip, SHA-256 checked before extraction, old install kept at `Programs\go-1.27.0` so rollback is a rename).

It was not the cause. All six archives still differed on the matched toolchain. The actual cause is that nothing used `-trimpath`, so every binary embedded the absolute paths of the machine that built it: a plain `yarg-sync` carried **906 occurrences of `C:/Users/<name>`**, 770 of the local `GOROOT` and 18 of the project directory. With `-trimpath`: zero, and 37,376 fewer bytes.

That is two problems, not one. Reproducibility was the one being chased; the other is that **artifacts this project mirrors publicly were shipping the builder's username and directory layout**. `-trimpath` is now on every build path — Makefile, the release job and the Dockerfile.

A note on how the first measurement lied: searching the binary for `C:\dev\YARG` returned **0** and nearly closed the question. Go normalises embedded paths to **forward** slashes; the same search for `C:/dev/YARG` returned 18. The size delta between the two builds was the thing that said "something is in there", and it was the only honest signal until the separator was fixed.

It took four causes, and each of the first three looked like the answer. The discipline that mattered was refusing to stop at "the archives differ, probably because X" and instead diffing the artifacts until they were identical:

#	Hypothesis	Verdict
1	Go version drift (CI 1.27.1, ENG-1 1.27.0)	Real, fixed — **not the cause**, all six still differed
2	No `-trimpath`, so absolute paths are embedded	Real, fixed — **not the cause either**
3	`vcs.modified=true` from a phantom-dirty tree	Real, fixed — made `yarg-sync.exe` identical, **but not** `yarg-song-server.exe`
4	CRLF reaching embedded content	**The rest of it** — see below

The fourth is the one worth the whole exercise, because it was never about checksums. `internal/httpapi/web/browse.html` is `go:embed`ed, and a per-extension `.gitattributes` rule had missed `.html` — so **the browse page the server serves had different bytes depending on which operating system built the binary**: 15,653 from Windows, 15,302 from Linux, same commit. `LICENSE` differed by 164 bytes for the same reason. Fixed with `* text=auto eol=lf`, with the testdata fixtures still excluded and their hashes verified unmoved.

Result, measured on `b90cc3e`: a Windows workstation and the Linux CI runner produced byte-identical archives for all six platforms. 6 match, 0 differ. Reproducibility is now a property this project can check rather than a sentence in a README. - **The launcher does not open a browser for you.** Launching a URL before the server is listening gives connection-refused, and the person then believes the download is broken when it is merely one second early. It prints the address instead. - **No GitLab Release object.** That needs `release-cli` on the runner host, which is the gitlab-ce session's box rather than this project's to change.

The determinism test had to be written twice, and the first one was green against broken code. It built twice in one process and compared bytes — which cannot catch `var epoch = time.Now()`, because a package-level var is evaluated once and both runs share it. It now asserts the stored timestamps against a date written in the test itself. Third time in this project that a test passed against the very defect it existed to catch.

Verified by doing it, not only in tests: twelve binaries built, packaged, the Windows zip extracted into a clean folder and `start-server.cmd` run as a double-click would. Server up, `GET /` 200 at 15,548 bytes, `/api/v1/features` and `/api/v1/library` both 200, version stamped, and the startup log naming the config file it read. Then the same thing again in CI.

Still not an app, and that is the honest description: it is a proper download. A tray icon and a service install are the next question. (The configuration menu that goal 1 asks for is now built — see phase 4 above.)

The whole path, run as a new user would, 2026-09-08

Every previous measurement had used a binary built on the machine doing the measuring. This one started from the published artifact and touched nothing else.

Step	Result
Download the CI artifact	7,773,966 bytes
Check it against the published `SHA256SUMS`	**match**
Unzip cold	six files, nothing scattered
Put 23 songs in `songs/`, double-click `start-server.cmd`	server up, `songs=23 problems=0`
`yarg-sync.exe -dry-run` from a **separate folder**	"would download 23", **0 files written**
`yarg-sync.exe` for real	23 songs, **238,215 bytes**, 0 failed
Run it again	**0 downloaded, 23 already present**
What landed	23 files, every one `<40 hex>.sng`, first bytes `SNGPKG`, **0 leftover** `.part`
Serve the SYNCED folder with a fresh server	`songs=23 distinct_charts=23 problems=0`
Compare chart hashes between the two servers	**identical sets, 23 vs 23**

Two things worth pulling out. **238,215 bytes is the same total four earlier sync clients pulled** and the same the fork's batchmode run reported — reproduced here by a server rebuilt from scratch weeks later, which is what deterministic packing was for. And the last two rows are the round trip: a loose folder packed to `.sng`, served over HTTP, written by a client, and rescanned by a different server still has **the same chart identity**. Identity survives the whole path, measured rather than assumed.

The launcher, the README's instructions, the config template and the checksum file were all exercised as written. Nothing in the archive needed correcting.

Phase 5 — LLM chart generation

The long-term goal, and the phase most likely to move. It depends on every phase above working.

- **Start from the tools that already exist.** Help:Charting names two, both free, that do the first half of this: **Ultimate Vocal Remover (UVR)** separates a mix into bass, drums, vocals and other stems, and Spotify's **Basic Pitch** generates pitched MIDI from a vocal stem — a baseline vocals chart to work from. The novel part of this phase is the instrument charting and the difficulty reduction, not stem separation, and building either from scratch would be waste.
- Upload any song; auto-generate instrument and vocal parts across all difficulty levels.
- Produce a complete, working, **editable** package — generated charts are a starting point, not a final answer.

- **Distribution constraint, non-negotiable:** the chart/vocal tracks must be packageable and distributable *separately from the audio*, so charts can be shared without the music.
- Note what YARN's guidelines establish about this: **visual synchronisation is itself a derivative work**, which is why they refuse no-derivatives licences. A generated chart is a derivative of the song it was generated from, not a neutral companion to it — so separating chart from audio reduces the problem, and does not by itself dissolve it.
- Runs against the existing GPU arbitration on the home estate rather than a new stack.

Client track (independent of the server line)

Can be picked up at any time; does not block the server.

- **Controller compatibility** — official Rock Band and Guitar Hero instruments first. Upstream handles this through PlasticBand-Unity, so fixes may belong there rather than in YARG.
- **Graphics** — general rendering and visual improvements.
- Both are ordinary upstream contributions: fork, branch off `dev`, PR to `dev`.

Remote queue — upstream issue #860, built 2026-09-09 (fork `225b03cb`)

#860 asks for "*remote Library Search/Queue management from a Webpage whilst YARG is running*" — karaoke, couch participation, parties. Open since August 2024, four hearts, and one comment pointing at a Discord proposal that adds up/down votes.

Most of it already existed, which is why the change is small. Measured before writing anything:

What #860 needs	What YARG already has
A queue	`Playlist` — `AddSong`, `RemoveSong`, `MoveSongUp`, `MoveSongDown`
A setlist being built	`MusicLibraryMenu.ShowPlaylist` (ephemeral), with Add/Start already in the UI
A queue during play	`GlobalVariables.State.ShowSongs` + `ShowIndex`, advanced by the pause menu *(replaced upstream — see the 11-02 note below)*
Search	`SortString.SearchStr`, the same normalised text the library matches on
A background network service	`DataStreamController`'s thread + cancellation pattern
Main-thread marshalling	`UnityMainThreadCallback.QueueEvent`, already trusted by the audio callbacks
An HTTP listener	**nothing — there is none anywhere in the shipped client**

So the build was the listener, the page, and the seams. Four new files, three public methods on `MusicLibraryMenu`, two settings lines. **No scene, prefab or asset changes**, which keeps the diff against upstream small if this ever becomes a PR.

`HttpListener`, and that was measured rather than reasoned about. On stock .NET for Windows it is backed by http.sys, where binding anything but loopback needs an administrator or a URL ACL — a trap that fails only on players' machines, after the feature looks finished. Unity ships Mono's managed implementation:

`Assets/Editor/HttpListenerProbe.cs` binds `127.0.0.1`, `+` and `*`, drives a real request through each, and all three pass with no elevation (Mono 6.13.0, Unity 6000.3.5f2).

The part that would have silently not worked: there are two queues. In the menu it is `ShowPlaylist`; once the setlist starts, gameplay runs off a *snapshot* in `State.ShowSongs`. Adding to the wrong one is indistinguishable from success to the guest — the request returns 200 and the song never plays. Adds are routed by what is actually happening, and when neither queue is reachable the song is held and flushed into the setlist as soon as the menu exists again. Edits to a running show refuse to touch the current song or anything behind it, because reordering those would leave `ShowIndex` pointing at a different song.

> **Superseded upstream, 2026-11-02 — there is ONE queue now.** Upstream's #1758 moved the live setlist behind a static `Setlist` owner and deleted `State.ShowSongs`, `ShowIndex` and `PlayingAShow`. `MusicLibraryMenu.ShowPlaylist` **is** `Setlist.Playlist`, so the two-queues hazard described above no longer exists and the routing that avoided it is gone. **The guard above is still right**, and for the reason given: the playing position is a position, so editing behind it repoints the show. The paragraph is left as written because it is why the feature was built the way it was, and because the hazard was real — it is the kind of claim that should be marked superseded rather than quietly edited into agreement with today's code. The port, and the 87 compile errors that forced it, are further down under *The fork was brought current on upstream `dev` by MERGE*.

Security follows this project's own `config_writes` shape: off by default; three values rather than a bool, because "on" would have had to mean *anybody on the wifi*; `Local` binds loopback so it is unreachable rather than reachable-and-refusing; locality comes from the socket's remote address and **never** from a header; and LAN mode refuses non-private addresses, so an accidental port-forward is a 403 rather than the internet queueing songs on somebody's TV.

17 checks over real HTTP in `Assets/Editor/RemoteQueueSmokeTest.cs`, red-proofed with one break — making `Local` bind `+` fails exactly one assertion, "*Local: the LAN address does NOT answer*", and nothing else.

The test found two real things. One was the harness: it called the requests from `EditorApplication.update` while the main-thread pump ran on the same callback, so the two marshalling endpoints deadlocked and looked broken — they were fine. The other is in the shipped code: this machine's only non-loopback address is Tailscale (`100.79.53.15`), which is CGNAT and not RFC1918, so LAN mode refused the player's own network. `100.64.0.0/10` is now allowed.

Not proven, and stated rather than left to be assumed:

- Batchmode has no menu scene and no song library, so **the two-queue routing is not exercised**. Until somebody plays it, "a queued song appears in the setlist" is inference from the code.
- The `HttpListener` result is the **editor**, which is Mono. A shipped player is IL2CPP with managed stripping — a different harness, and it needs its own measurement.
- The smoke test pumps `UnityMainThreadCallback`'s queue from the editor loop rather than from `MonoBehaviour.Update`, so a defect that only appears under the real update loop would not show.

This is a fork prototype, not a PR. Upstream's `CONTRIBUTING.md` says to raise a feature on Discord first, and the tier decides whether a PR is read at all; #860's comment points at a Feb 2025 Discord proposal, so somebody may already be building it. Asking is the next step, and now there is something to show — which was Jay's condition for making contact.

Increment 3 — the room votes (fork `dc46f591`)

860's one comment points at a Discord proposal that "*adds up/down votes to the queue*". This is

that, plus the half Jay asked for: **anybody can put a song forward from the library, and the room decides whether it gets in at all.**

Two stages, because suggesting and queueing are different things:

1. Somebody searches and **suggests**. It does not enter the queue. Suggesting counts as the suggester's own upvote — nobody suggests a song they would not vote for, and making them tap twice just produces a board of zero-score suggestions.
2. At the threshold it is **promoted**, and a **second vote** decides where it lands: play next, or on the end of the setlist. First side to the threshold wins.

First-to-threshold rather than a countdown. A timer means a clock running against game state, and "*we are still waiting for the vote to close*" is a worse party than "*three people tapped, it is happening*".

Play next, not play now. The request said "play it now"; this plays it *next*. Cutting off whoever is mid-song is not a party feature. That is a deliberate reading of the request, and a one-line change if it is the wrong one.

Songs already queued can be voted on, and the unplayed queue reorders by score. **The sort is stable, and that is load-bearing**: with no votes every score is zero, so the order comes back untouched and a hand-arranged queue survives until somebody actually votes. Only the unplayed part moves — during a show everything up to and including `ShowIndex` is left alone, because `ShowIndex` is a position rather than a reference.

Voting state touches none of YARG's data model — no `Playlist` change, no save format, nothing persisted; a restart starts a fresh room. Small diff, and a party's argument does not outlive the party. Identity is a random id the page keeps in the browser: enough to stop a double-tap and one phone voting twenty times, not enough to stop somebody clearing storage — which is the right amount of ceremony for a living room.

33 checks, 0 failures. The 12 new ones are the valuable ones: unlike the queue plumbing, the voting rules are plain data with no Unity in them, so batchmode can check them properly — one voter cannot vote twice, changing your mind moves your vote, the threshold promotes exactly once, the second vote settles only at the threshold, switching sides moves the vote, and the sort is stable without votes and ordered with them.

The test found a real defect again. Every voting endpoint was a 500 in batchmode because it read `SettingsManager.Settings` unguarded — and the mode callback deliberately runs when a *saved* setting is loaded, so the listener really can be answering requests before settings exist. It now falls back to defaults.

The queue plumbing these votes drive — insert-at-top, reorder, play-next during a show — is still **inference**, for the same reason as increment 1: no menu scene and no library in batchmode.

Increment 4 — the vote needs a moment to happen in (fork `6bfae300`)

Jay's observation, and it undercut an argument from increment 3: **while a song is playing, everybody's hands are on an instrument.** Nobody is holding a phone. So voting is not a continuous background activity — it happens in the gap between songs or it does not happen. Which also means a countdown belongs here after all; a vote with a dedicated window wants a clock, where a vote running under a song did not.

What it turned out not to need, measured rather than assumed. The instruction was to stop the setlist progressing until the vote concludes. **YARG does not progress on its own** — `ScoreScreenMenu` advances only when a human presses Green, with no auto-advance and no timer anywhere in it. There is nothing automatic to block.

What there *is* to prevent is the habitual press: somebody hits Green from muscle memory before anyone has voted, and the suggestion is silently discarded. So while a vote is open the button reads *"Voting… (press to settle)"*, **the first press settles the vote and redraws rather than advancing**, the second press is an ordinary Continue, and if nobody presses anything the window closes itself after 30 s. Two independent ways out, so a room where somebody wandered off cannot be stuck — and the console keeps the last word, because it is somebody's living room.

On expiry the leading suggestion wins if the room actually wanted it. One already at the second vote outranks a higher-scoring suggestion that has not got there yet — the room has spoken about it once — and its own play-next/setlist lead decides where it lands, a tie going to the setlist as the less disruptive option. Suggestions nobody wanted are dropped; **everything else survives to the next gap**, because throwing away cast votes teaches people not to bother.

The insert point needed no change. At the score screen `ShowIndex` is still the song just played, and Continue does `ShowIndex++` before loading — so inserting at `ShowIndex + 1`, which a won play-next vote already did, is exactly "the song that plays when Continue is pressed".

Still no prefab or scene changes: the button label is a `NavigationScheme` entry, so the voting state is one new key in `en-US.json`, and `Update()` goes on a `MonoBehaviour` that already exists.

46 checks, 0 failures. One defect fixed before committing rather than after: `ShouldHold()` was calling `Suggestions()`, which builds and sorts a list — from `Update`, sixty times a second for the length of the window. It now asks a count.

The hook itself — does the button really relabel, does the first press really settle without advancing — is **inference** until somebody plays a setlist. Batchmode has no score screen.

Increment 5 — a real phone app (new repo `yarg-remote`, `a18c3b8`)

Jay: *"build an iOS and Android app instead of the web app"*, then *"check how we did this for the ShopStack app"*. So it follows the house pattern that is already in daily use on this workstation: **Capacitor 8 + React 18 + TypeScript + Vite**, a bundled web UI inside a native shell, with the server address typed into the app once and remembered.

"Instead of the web app" turned out to be less of a rewrite than it sounds, because Capacitor *wraps* a web app — the built-in page the server already serves stays exactly where it is, for anyone who would rather open a browser.

GitLab	`fatalexception/yarg-remote`, project 57
Mirror	`github.com/Coffeehedake/yarg-remote`, GitLab mirror 13, verified with `git ls-remote`

Tabs	Find / Up next / Suggestions, matching the built-in page
Tests	21 unit tests green; `tsc` and `vite build` clean

Two things are load-bearing and neither is visible at a call site. `X-Yarg-Remote` goes on *every* request including reads — losing it would break every write while every read kept working, which is the worst possible failure to debug on a phone at a party, so a test asserts it. And `CapacitorHttp` routes `fetch()` through the native stack, so requests carry `capacitor://localhost` or `http://localhost` as their origin; the server allows exactly those (`IsNativeShellOrigin`), and that allowance is what makes the app work at all.

Discovery is a follow-on, not a prerequisite — a claim this session made and then withdrew. The argument was that the server needs mDNS before an app is usable. ShopStack Mobile has **no discovery at all**: its server URL is typed into Settings, and it is in daily use. Manual entry is the proven house pattern; mDNS makes the connect screen optional and nothing more.

The wire contract is MEASURED, not inferred. `Editor.RemoteQueueHost` in the fork (`e85d2ffe`) holds a real listener up in batchmode; `yarg-remote/src/live.test.ts` drives the app's own client against it. **12 cases, 12 pass.** They are written to distinguish *why* a call failed, because the status code alone does not say it:

The case	What a failure would have meant
`POST /api/queue` → 404 **"no song with that hash"**	a wrong path gives 404 *"no such endpoint"*; a wrong method 405; a missing mark header, refusal before the hash is read
`DELETE` must not give 400 *"'a 'hash' is required"*	the hash went in the body, where HttpListener cannot see it
`?on=queued` → 404	the suggestion branch answers 200 for anything, so a 200 means the parameter never arrived
`?choice=sideways` → 400	proves the choice parameter is read rather than defaulted
a mutating request with no `X-Yarg-Remote` is refused	the whole CSRF story

Red-proofed both ways. Against a dead port every case fails; breaking one route spelling in `api.ts` (`/api/queue` → `/api/queues`) turns exactly the queue cases red with *"no such endpoint"* instead of *"hash"*.

And the first green was partly false, which is the part worth keeping. The dead-port run initially left **2 of 12 passing**: both asserted only `status !== 400` , and an unreachable console is `status 0` , which is not 400. A wrong answer and *no answer* satisfied the same assertion. Both now require a real answer, and the dead-port run is 12 red.

What the contract test still cannot prove: batchmode has no song library, so every hash is unknown and nothing can actually be queued. It settles that we ask the right questions at the right addresses and read the answers correctly. *"Queueing a real song works"* remains inference until somebody plays the game — the same gap increments 2–4 already carry.

What is NOT verified:

- **No `cap add` has been run.** iOS and Android are untested and the two cleartext exceptions (`NSAllowsLocalNetworking`; an Android network-security-config) are written down in the repo's README rather than applied.
- Codemagic is not wired up.

The fork was rebased onto upstream `dev`, 2026-10-30

`yarg`'s `dev` had fallen 110 commits behind `YARC-Official/YARG`'s and carried 26 of ours. It was rebased (`dev` is now `92c3f007`: upstream's tip plus our 26, **0 behind**) and force-pushed to GitLab; the mirror carried it to GitHub. Staying abreast of upstream is the point — an upstream PR from a fork that has drifted this far is a rebase first and a PR second.

- **Two of the 26 conflicted.** The first (the song-server sync commit) was adjacency: both sides added a method at the same spot in `SettingsManager.Settings.cs`. The seventh (the server status row) was a real API change: upstream replaced `AbstractMetadata`'s `isAdvanced` flag with a `visibleWhen` delegate, so our `TextMetadata` constructors lost the parameter. Nothing else of ours used it.
- **Every fork commit hash in these docs changed**, because a rebase rewrites them. They have been remapped to the new hashes, matched by commit subject. **A measurement quoted "at" a commit was taken on the old base** — the pre-rebase tip is kept, under its old hashes, as the branch `dev-pre-rebase-2026-10-30` on GitLab and GitHub.
- **Anyone with an older clone of `yarg` must `git fetch` and `git reset --hard origin/dev`** — a plain `git pull` will not do the right thing, and local work goes on a branch first.
- **Upstream's LFS objects were not on our GitLab.** A checkout of `upstream/dev` died on `NotoSansSymbols2-Regular.ttf` with a 404 until `git lfs fetch upstream upstream/dev` fetched them and the push uploaded 8 of them to ours. Expect the same after any later rebase that pulls in new LFS assets.

Measured on the rebased tree: the fork compiles under Unity 6000.3.5f2 headless — **0 `error CS`, exit 0**; the yarg-remote live contract test passes **12 of 12** against a `RemoteQueueHost` built from it; `YARG.Core` is **737 passed / 6 failed / 2 skipped of 745** (the six are upstream lip-sync phoneme tests, see `WORKSPACE.md`).

Played the same day — what that found. The rebased fork was built and run, and the playtest earned its keep:

- **The Song Server tab crashed the game on first click**, a stack overflow, not an exception. With no URL set (every fresh install) entering the tab redrew the menu, the redraw asked the status row for its text, the row started a refresh, and a refresh with a blank URL "settled" by redrawing — forever. It was **our bug, not the rebase's**: the files involved are identical before and after. Every earlier test ran with a URL already configured, and the headless probe had no settings menu to redraw, so it tested the row alone. Fixed in `yarg` `93e48422`; the probe now stands in for the menu through a `Redrawn` event, and the two new checks were seen to FAIL on the old code (22 redraws before the probe cut it off) before they passed.
- **"Insecure connection not allowed" against a healthy server** — a *release* build refuses `http://` because `insecureHttpOption` is `DevelopmentOnly` on purpose (ADR-004). Not a regression. The build script takes `YARG_BUILD_DEVELOPMENT=1` now (`282755b2`); see `BUILD.md`.
- **Sync against the live server mirrored 127 of 128 songs** (1,188 MB, ~80 s, every file matching a catalogue entry). The missing one, *Sepultura – Roots Bloody Roots*, was **a server defect**: the `.sng` writer rejected a `;` in a metadata value (`charter = TheSheepQueen; Whizzer`), so `GET /song/<hash>.sng` returned HTTP 500. The rule was an unverified claim about the format — the reference encoder packs and decodes that value unchanged. Fixed in the

writer and pinned with a SngCli fixture, released as `v0.1.2` and **deployed to vault2 the same evening** — see `DEPLOY-VAULT2.md`, *Fifth deployment*. The game then synced the one missing song (`downloaded=1 failed=0` , 2,774,330 bytes, the same SHA-256 the server serves), and **YARG's own song cache holds its title, artist and the full `charter = TheSheepQueen; Whizzer`** , with no `badsongs.txt` . So the real game does accept a `;` package. (The dev build on the rebased base, not the installed v0.15.0 oracle.)

- **The remote queue (#860) against the running game, 12 of 12.** The contract test, which until then had only run against a headless host with no songs, passes against a real game with the 128-song library. Measured through the API the phone app uses: the library lists all 128; adding by hash works; adding the same hash twice leaves one entry; an unknown hash is a 404 "no song with that hash"; remove works. **Where a song lands depends on a screen:** with the game on Settings every add went to `target: "pending"` , and a *move* returned 409 "that song cannot be moved from where it is" — because the setlist lives in the music-library menu and a pending song cannot be moved. With the music library open the target is `setlist` , three songs queued in order, and moving one down reordered them. The game's own `setListNotEmpty` flag flipped on screen (the button went from "Play Song" to "Add to Setlist / (Hold) Start the Set"), which is the game confirming its setlist received them. That 409's wording blamed the song when the cause was the closed menu; **fixed in `yarg` `61d2bc55`** — `Move` now says why, with the same status and the original wording kept for the other refusals (checked on a rebuilt game: a queued song with the library closed names the closed library; a never-queued real song and an unknown hash do not). **Checking that fix turned up a worse bug, also fixed:** with the library *open*, `Move` answered `200 moved: true` for moves that moved nothing, and for a song that was never queued, with a non-empty setlist, **moving it down was an HTTP 500** — upstream's `Playlist.MoveSongDown` does `IndexOf` (−1) and then indexes `SongHashes[-1]` . `Move` returned success unconditionally and `MoveSongInSetlistRemotely` returns nothing, so nothing checked. It predates the message fix; the server survived (the exception was caught, the queue stayed intact) but any client could trigger it. `Move` now looks first: a song not in the setlist, or already first (up) or last (down), is the usual 409. **Measured in the game across every case:** never-queued down and up, on an empty and a non-empty setlist, all 409 (were 500 and 200); first song up and last song down 409 (were 200 no-ops); middle moves still 200 and reorder. **Correction to an earlier version of this paragraph:** it said fixing this would change what the phone app sees, an up-arrow on the first song erroring. That was written without reading the app, which has no move controls and never calls `move` — its reordering is `voteQueued` . The endpoint's only caller is the contract test, so nothing user-facing depended on it, and the cost of leaving it was a reachable 500, not a broken screen.

- **The Remote Queue settings showed raw localization keys** (`SETTINGS.SETTING.REMOTEQUEUE.NAME`): `en-US.json` never had entries for `RemoteQueue` , `RemoteQueueVoting` or `RemoteQueueVotesNeeded` — at `7f17f910` and in upstream too, so not a rebase artefact. A missing key fails nowhere, which is why it survived two playtests. Fixed in `yarg` `d9474604` , and `SettingsRowProbe` now requires a name and a description for every setting and a `Dropdown` block for every enum dropdown (172 settings; it was seen to fail on exactly those three first). **What it cannot see**, stated in its source: an offered dropdown option whose string is missing when the block holds others, because that needs a live settings object and constructing one in batchmode throws (no audio manager). Confirmed on screen afterwards.

- **Suggestions and voting, run end to end in the real game** (library screen open, threshold 3). A voter is the socket's own address, so three voters were simulated from `127.0.0.1` , `.2` and `.3` — Windows treats all of `127.0.0.0/8` as loopback, and the server sees three different addresses. A suggestion starts at score 1 (the suggester); **six votes from one address, each with a different `X-Voter-Id` header, left the score at 1** — the 51-votes-against-a-threshold-of-3 problem that motivated using the socket address holds in the game, not only in the unit-tested rules. A second and third address took it to 3, promoted it to `deciding` , and three `choice=setlist` votes ended

`decided: true, outcome: AddToSetlist` with the song in the setlist and off the board. A second song put through `choice=next` ended `PlayNext` and **landed at the head of the setlist**. Votes on queued songs (`?on=queued`) reorder by score: one down-vote moved a song below its neighbour at once, and the score kept falling to −3. All state is in memory and was cleaned up (queue and board empty). **PlayNext was measured with nothing playing**, so "straight after the song currently playing" is untested; and an address is not a person — two people sharing a tablet get one vote, which the code says.

- **The phone app's own UI, driven in a browser against the running game — not a phone.** `yarg-remote` runs unmodified under Vite, and a **test-only, untracked dev proxy** stands in for what the Capacitor shell does natively: it makes the browser's calls same-origin (the game sends no CORS headers) and relabels them `Origin: http://localhost`, which the game's origin gate accepts for a native shell. The proxy was checked first with a POST carrying a browser-style origin, then deleted. What the app did, with the game asked separately each time: it connected and listed the library (**"Showing 100 of 128"**, `;` charter included); search narrowed; **Queue** put the song in the game's setlist and the tab became "Up next (1)"; the vote arrow moved the game's score to 1; **Suggest** put it on the board ("Suggestions (1)"); after two other voters the app moved on its own poll to **"WON - NOW: NEXT, OR THE SETLIST? (3 DECIDES)"**; its "Add to setlist" press registered at the game (`for_setlist=1`); remove worked through `DELETE`-with-query; the empty states read correctly; a wrong address gives "*Nothing answered at http://... Check the phone is on the same Wi-Fi, and that Remote queue is set to LAN*", is not remembered, and a reload reconnects to the last good one. **Two defects found.** (1) The Suggestions note said "*A song is playing … nothing moves until the score screen*" whenever `intermission` was false, which includes a game idling on a menu, and a decided vote moved the song at once — **fixed** in `yarg-remote` (reworded to what is established). (2) **The app and the README both say the game "will show an address" when Remote queue is on. It did not**: the game wrote only `Remote queue: listening on http://+:8099/` to its log (a wildcard, not an IP) and showed nothing in any menu or toast, so a first-time guest had no way to learn what to type. **Fixed in the game**: a live status row under the Remote Queue setting (Settings → Experimental) says what is true — off; "*Only this computer can reach it: 127.0.0.1:8099*" for Local; for LAN "*Type this into the app: 192.168.2.46:8099*" followed by fallbacks, the address on the interface with a **gateway first** (here Wi-Fi) ahead of Tailscale, the VPN and the Hyper-V/WSL switches, never loopback or link-local; and, when a start fails, "*Could not start (Lan): *" — a failed start used to be a log line and nothing else, leaving a setting that said LAN over a server that was not running. The sentences are pure and checked headless, and the checks were **mutation-proofed** (break a rule, watch the matching check go red, restore). **Seen on screen** in Local and in LAN. Independent of the row, the game was listening on `0.0.0.0:8099` and **answered** `200` **on** `192.168.2.46:8099`, i.e. from its LAN address and not loopback. **A mistake worth keeping:** I first reported that LAN mode cannot bind for an ordinary Windows user (and wrote code suggesting a `netsh http add urlacl` command). That came from a test in PowerShell 7, where `HttpListener` is HTTP.sys and does need a URL reservation; the game is Unity's Mono, whose `HttpListener` is managed and does not. The running game, not elevated, bound LAN fine. The wrong runtime was measured; the claim and the command were removed before anything was committed.

Still not verified: the app on an actual phone — no iOS or Android build has been made, so the native HTTP stack, the cleartext exceptions, touch and layout on a real screen are all untried; what the game's own setlist *shows* (only that it says it is non-empty); playing a queued song, and so PlayNext while one is playing; the `voting is off` 404 (needs the setting flipped); and votes from real separate devices rather than loopback aliases.

The fork was brought current on upstream `dev` by MERGE, 2026-11-02

`yarg`'s `dev` was 12 commits behind `YARC-Official/YARG` and 41 ahead. It was **merged**, not rebased — `dev` is now `b8329d70`, the merge commit itself `301ccc0e`, **0 behind** — and pushed to GitLab with no force.

Why a merge, when 2026-10-30 was a rebase. That rebase was right at 110 behind: the drift was large enough that replaying our work onto upstream's tip was the only honest way to see it. At 12 the arithmetic flips. A rebase rewrites all 41 of our hashes, and those hashes are quoted all over these docs, so the cost is not the replay but remapping the written record by subject afterwards — and a hash in a doc that silently names nothing is worse than a merge commit in a fork's history. The upstreaming argument does not apply either: the documented PR flow creates a branch on GitLab and opens the PR from the mirror, so a PR gets its clean base when it is made.

`dev-pre-merge-2026-11-02` holds the pre-merge tip. A merge needs no such branch; it is kept for symmetry with `dev-pre-rebase-2026-10-30`.

`git merge-tree` said "no conflicts" and the build was broken in 87 places

Checked before merging, with `git merge-tree --write-tree --name-only HEAD upstream/dev`: clean, no conflicts. That was true, and it was about TEXT. The merged tree then failed to compile with **87** `error CS`.

Upstream's **#1758, "Move the live setlist into a single Setlist owner"**, deleted `PersistentState.ShowSongs`, `ShowIndex` and `PlayingAShow` and put the live setlist behind a static `Setlist`. Our remote queue (#860) was built on all three: 78 of the errors are in `RemoteQueueBridge.cs` and 9 in `ScoreScreenMenu.cs`. **`RemoteQueueBridge.cs` was not in the conflict surface at all** — git had nothing to report, because neither side edited that file; upstream changed the *type it depends on*. Seven files were touched by both sides and every one of them merged cleanly.

So: a clean merge is a statement about text, and the thing worth measuring is a build. The compile is what found this, and it found it in 60 seconds.

What the port had to decide, not just rename

The three deleted members do not map one-to-one onto the new owner, and two of the differences change behaviour rather than spelling:

- **`ShowSongs` was a mutable list; `Setlist.ResolveSongs()` allocates a NEW one each call.** Translating `ShowSongs.Add(song)` to `ResolveSongs().Add(song)` compiles, runs, and does nothing — the add lands on a temporary. Mutations go through `Setlist.Playlist` instead, whose `AddSong` / `RemoveSong` / `MoveSongUp` / `MoveSongDown` raise `Edited`; `Setlist` subscribes to that and re-raises it as `Changed`, which `MusicLibraryMenu` now listens to. Upstream documents the trap in the class itself: *"Writes straight to `SongHashes` don't raise it."*
- **`ShowIndex` counted positions in the snapshot; `Setlist.Index` counts positions in `ResolveSongs()`, which SKIPS hashes the library cannot resolve.** So `Index` and a position in `Playlist.SongHashes` disagree by one for every unresolvable hash ahead of the playing song. Guarding with one and editing with the other is how an edit lands on the wrong side of the playing position — and it would do so only on a library where a queued song had gone missing, which is to say never on the machine it was written on. The whole show path is in hash space now, via a `PlayingHashIndex()` that maps across once.

- `Playlist` **has no insert-at-position and no partial sort**, which "play next" and a settled vote both need. Doing those as `Clear()` plus a run of `AddSong` is what the menu helpers already did, and it is fine on a setlist nobody is playing: during a show it empties the list the playback position points into and raises an `Edited` per song. `Playlist.ReplaceSongs` was added for that — one step, one event, no intermediate state.

The feature's central hazard is gone. `RemoteQueueBridge`'s own summary used to open with "*THERE ARE TWO QUEUES*", and the section above in this document says the same. Upstream's #1758 removed the duality: `MusicLibraryMenu.ShowPlaylist` **is** `Setlist.Playlist` now. What survives is the guard, which was never an artefact of having two queues — the playing position is a position, so editing behind it silently repoints the show.

`_pending` **is now a choice rather than a necessity**, and was kept deliberately. It existed because reaching the setlist needed a `MusicLibraryMenu` instance; `Setlist.Playlist` is static and reachable with no menu at all, so a song never has to be held. It stays because `pending` is part of the HTTP contract `yarg-remote/src/types.ts` transcribes, and widening a currency merge into a behaviour change is a different change. **The HTTP surface did not move**: no string literal the phone app reads was touched, so the app needs no update.

A compile cannot tell a correct offset from one that is out by one

The port rewrote the rule that protects a running show from the phones, in a different index space, and nothing tested it. `RemoteQueueSmokeTest` covers the HTTP surface and the voting, not the setlist mutation paths, because those need a show — and a show cannot be started headless: `Setlist.TryStart` needs songs that resolve, and batchmode has no audio backend, so most of the corpus will not scan. Asking for a real show would have meant this went unmeasured.

So the arithmetic became the unit. `RemoteQueueBridge` exposes `CanEditWithinShow`, `CanMoveWithinShow`, `QueueWithSongNext` and `QueueInVoteOrder` as pure functions over plain numbers, and `Assets/Editor/SetlistGuardProbe.cs` measures them — 26 checks, no library, no scene, no show. Same shape as the library badge, which became measurable the same way.

They are `public`**, not** `internal`**, and that was measured rather than chosen.** `Assets/Editor` is a separate assembly, so `internal` does not reach the probe: the first run died with **81** `error CS0117` naming every one of them. "Make it internal so the test can see it" is simply false in a Unity project.

A second break the compiler could not see, found by reading the overlap

Seven files were changed by both sides. All seven merged cleanly, and the build is green, so nothing in the tooling had anything left to say about them. Reading one of them anyway turned up a real defect in **our** code:

Our remote helpers on `MusicLibraryMenu` end with `if (ShowPlaylist.Count == 1) SetNavigationScheme(true);` — rebuild the green button's navigation scheme when the setlist stops being empty. Upstream now does that itself, in `OnSetlistChanged`, subscribed to `Setlist.Changed` — **and it checks** `IsNavigationSchemeBlocked()` **first**, deferring the rebuild through `_needsNavigationSchemeRefresh` when a popup menu or a dialog is showing.

Ours did not check. A phone can queue a song at any moment, including while the player has a popup or a dialog open on the TV, so the remote path could rebuild the navigation scheme out from under an open popup — which is precisely what upstream's guard exists to prevent. Our calls were removed rather than guarded: the subscriber already handles the transition, and correctly.

Not found by the compiler, not found by the probe, and not reported by `git`. Found by asking what the other side had changed in a file that merged without complaint.

Proved able to go red before its green was believed, once per function. The prediction was written down before each run and compared with the measurement; all three matched.

Break	Predicted	Measured
`CanEditWithinShow` as `at >= floor`	1 failure: *the song being played cannot be removed*	FAIL (1), that check
`QueueWithSongNext` without `RemoveAt(existing)`	1 failure: *and is not left in its old place as well*	FAIL (1), that check, reporting `[0,1,4,2,3,4]`
`QueueInVoteOrder` without the put-back loop	2 failures: the partial sort and the unknown-song sort	FAIL (2), those two

Trying to make the probe catch a fourth break is what found dead code. The first plan for break 2 was to delete an `if (existing < at) at--;` that adjusted the insertion point for the position a moved song vacates. Working out which check would notice showed that **none could**: the branch is reached only when `CanEditWithinShow` says yes, so `existing >= floor + 1`, while `at` is at most `floor + 1` — the song being moved is never before the slot it moves to, so the removal never shifts that slot. The arithmetic could not execute. It was removed rather than commented, because code that cannot run still has to be read, and it claims a case exists.

That is the second time on this project that an attempt to verify something found the instrument or the subject to be the problem rather than the thing under test. It is cheaper than the first.

Measured on the merged and ported tree

- **The fork compiles under Unity 6000.3.5f2 headless — 0 `error CS`, exit 0.** The same command reported 87 before the port, which is the only reason anyone knew there was a port to do.
- **`SetlistGuardProbe`: 25 of 25.** Red three ways first, as above.
- **`YARG.Core`: 750 tests, 742 passed, 6 failed, 2 skipped**, the stable result on four of five runs. The submodule pointer moved `8ae7718` → `6c0923e`, so the baseline written down in `WORKSPACE.md` had to be re-measured rather than carried forward: a changed pointer means the old numbers describe different code. Detail, including the intermittent seventh failure, below.
- **The yarg-remote live contract test: 13 of 13**, against a headless `RemoteQueueHost` built from the ported fork. This is what covers the HTTP paths the probe does not: the probe measures the show-guard arithmetic with no game, this drives the real endpoints.

The YARG.Core baseline moved, and one of its tests is a wall-clock race

The submodule pointer moved `8ae7718` → `6c0923e` (upstream's #478, expose chart file modified time, and #471, editable vocal colors), so the baseline in `WORKSPACE.md` describes different code and was re-measured. **New baseline: 750 tests, 742 passed, 6 failed, 2 skipped.** Previously 745 / 737 / 6 / 2, so five tests were added and all five pass; the six failures are the same lip-sync phoneme tests, and still upstream's — our submodule pointer equals `upstream/dev`'s exactly and none of our commits touch `YARG.Core`.

A seventh failure appeared once in five runs and is worth writing down rather than averaging away: `PreviewContextTests.WaitForCompletionAsync_AllowsFadeOutAfterCancellation`. It is a race against the wall clock, not a flake of unknown cause — the test creates a context with `fadeDuration: 0.05`, cancels, `await Task.Delay(10)`, and then asserts the wait task is **not** complete. Any scheduling hiccup longer than about 40 ms between those two lines lets the 50 ms fade finish and the assertion flips. It failed on the one run that shared the machine with other work and passed on the four that did not.

It is upstream's test and upstream's race, and it is a candidate for `UPSTREAM.md`: unlike the `song.ini` bounds check looked at on 2026-11-02, where our own reading was wrong, this one is demonstrable from the test body alone.

An oddity, recorded rather than explained: four tests print as `Skipped` while the summary counts 2, and 742 + 6 + 2 = 750 adds up. Two of them (`Canonical20260801_*`) are evidently not counted in the total. Not investigated; noted so a later reader does not think the arithmetic is broken.

What is NOT verified *(at the time this was written — it was played the same day, see the playtest entry below)*. None of this was played. The probe measures arithmetic and the contract test drives HTTP against a headless host with no songs; nobody has queued a song from a phone during a real setlist on the ported build, which is the only thing that would exercise the guard end to end. The 10-30 entry above is the precedent for why that matters: the rebased tree passed every headless check and then crashed on the first click of the Song Server tab. **The twelve upstream commits this merge brought in — editable vocal colors, vocal track fading, the solo and unison box fix, notification history, Clone Hero practice-section names, Sort by Date Added, the VLC update — are likewise unplayed.**

Played, 2026-11-02 — and the playtest paid for itself twice before a song started

The 10-30 entry is the precedent: a tree that passed every headless check crashed on the first click of the Song Server tab. This one did not crash, and still found more than the headless work did — including one thing that started downloading a server.

The build. 603 MB in 2.3 minutes, `PASS WITH 2 ERROR(S)`, both `RenderTexture.Create failed` from `-nographics` and both the class `BUILD.md` already documents. All ten of the ported symbols are present in `Assembly-CSharp.dll` (`CanEditWithinShow`, `CanMoveWithinShow`, `QueueWithSongNext`, `QueueInVoteOrder`, `ReplaceSongs`, `PlayingHashIndex`, `InsertNextInShow`, `WithServerBadge`, `BadgeMarkup`, `IsFromServer`), alongside upstream's new `ToastHistoryTab`, `PracticeSectionHelper` and `TryAdvance`. So this is a build of the merged tree, not a stock one.

A "development build" does NOT use the `dev` data folder, and that started a 3794-song sync

`YARG_BUILD_DEVELOPMENT=1` sets `BuildOptions.Development`, which is what makes `insecureHttpOption` allow plain http. It does **not** define `YARG_TEST_BUILD`, and `PathHelper.Init` reads:

```
#if UNITY_EDITOR || YARG_TEST_BUILD   -> persistentDataPath/dev
#elif YARG_NIGHTLY_BUILD              -> persistentDataPath/nightly
#else                                 -> persistentDataPath/release
```

So the headless build lands in `release/` — **the same folder a previous release build left its settings in.** Settings written into `dev/` for this playtest were never read. What the player read instead was the 2026-10-30 configuration: `SongServerUrl` pointing at vault2, and `SyncOnStartup` on. It began mirroring the server unattended, and the server

now holds **3794** songs rather than the 128 it had in October. It reached 251 new files and about 7 GB before it was killed. The loading screen read *"Downloading song 58 of 3794"*; Jay saw it and said so, which is the only reason it was caught in minutes rather than when the disk filled.

Use `-persistent-data-path <dir>` to isolate a playtest. `-offline` is also honoured. Both are upstream's, in `CommandLineArgs`.

Trimmed back the same evening, on Jay's call. The split was clean enough to act on without guessing: exactly **128** `.sng` files dated 2026-10-30 (the original sync, and the number this document already recorded) and exactly **250** dated 2026-11-02, plus the one `.sng.part`. The 251 were listed to a file and counted BEFORE anything was deleted, then removed by literal path one at a time — not by a glob and not by `-Filter`, because this project has a measured incident where `Get-ChildItem -Filter "*.go.*"` matched `api.go` and deleted every `.go` file in a directory. Result: 128 files, 1.2 GB, all 2026-10-30, and the 10-30 `songcache.bin` still describes exactly them. Nothing else in the folder was touched.

`SyncOnStartup` **defaults to ON and has no ceiling.** It does not ask, it does not cap, and the only sign is one line on a loading screen. Pointed at a large server on a fresh machine that is potentially hundreds of gigabytes before the player reaches a menu. The setting is doing exactly what it says; the question is whether "mirror everything, silently, on every start" is the right default, and it is a product decision rather than a bug. Worth settling before a public beta.

The sync is interrupt-safe, and that was measured rather than hoped. Killing it mid-download left exactly one `10874c46….sng.part` and no truncated `.sng`. The scanner ignored the partial and read 378 of 378 real songs, which is also why the file count and the song count differ by one.

Two things that are not the merge's doing

- **Upstream's asset CDN does not exist.** `yarg-assets-release.ulna.net` is NXDOMAIN while the rest of the internet resolves, so every launch logs two content-catalog exceptions. Identical in the 2026-11-01 logs, so it predates this work entirely.
- **The first-run modals block the song scan.** With `ShowExperimentalWarningDialog` or `ShowEngineInconsistencyDialog` still true, the loading sequence stops in front of them and the scan never starts. The log ends at *"Initialized phoneme dictionary"* and the game looks hung while it is simply waiting for a click.

The ported remote queue, measured against the real game

378 songs scanned in **1.26 s**. The remote-queue listener binds in a real Mono player, which narrows the KNOWN GAP in `BUILD.md` that said the bind had only been measured in the editor.

What	Result
Songs queued from the main menu	`target: "pending"`, held
Opening the music library	log: `flushed 3 pending song(s) into the setlist`, target → `setlist`
Move up, twice	order changed both times
Move up from the front	`409 "that song cannot be moved from where it is"`, order untouched
Six adds then six removes, back to back	12 × `200`, ~0.1 s each, setlist ends empty

And with a show actually running — the part no headless check could reach, because `Setlist.TryStart` needs resolvable songs and batchmode has no audio backend:

Guard	Result
Remove the song being played	`409`, still in the queue
Move the next song up into the playing slot	`409`, still at its index
Remove a song further ahead	`200`, gone
A won "play next" vote	landed at index 1, **immediately after** the playing song; the songs behind it shifted right; the playing song and index unchanged

The play-next case is the one that matters most, because `QueueWithSongNext` and `Playlist.ReplaceSongs` were written for this merge and had only ever been exercised as pure functions by `SetlistGuardProbe`. Driving it needed three voters, and a voter is the request's remote address — so the three votes were sent from `127.0.0.1`, `127.0.0.2` and `127.0.0.3`, which is three distinct voters out of one machine and did not require reconfiguring the running game. The nomination promoted on the third vote and decided on the third choice.

The instrument was broken twice, and neither was the game

- **HTTP 000 from five of six removes**, with the songs still in the setlist. It looked exactly like a defect in the ported remove path, and the next step was to go and find it. The timing stopped that: the failures returned in **nineteen microseconds**. Nothing had been sent. A curl process per request from a bash loop was failing to start. Re-run from a single process, all twelve requests pass. "*A failing measurement is not automatically a failing system*" now has a fourth instance, and the only reason it was caught is that the number was absurd rather than merely bad.
- **A wrong expectation printed next to a passing check.** The show-guard script announced it expected a plain add to land immediately after the playing song. It landed at the end, which is correct — `Add` appends, and landing next is the *play-next* behaviour. The check that actually asserted anything was about the playing song not moving, so the run reported PASS with a false sentence above it. A probe that prints its expectation must have that expectation checked by something, or it is just a comment that can rot in place.

Not covered

Nobody played a note. No instrument was connected, no song was played through, and the twelve upstream commits this merge brought in — editable vocal colors, vocal track fading, the solo and unison box fix, notification history, Clone Hero practice-section names, Sort by Date Added, the VLC update — were not exercised beyond their code being present in the binary. The per-song server badge was not re-checked on screen either: the playtest library was pointed at as an ordinary song folder, so no song in it was mirrored for this build. The badge was measured headless twice the same day.

A finished feature nobody could switch on — 2026-11-02 (`yarg` `55eda66d`)

Asked to build ADR-004 increment 2, *fetch a song when it is played*. It had been built since `36ca1f2a`. **The third stale status line in one day**, and the first one caught by the rule the other two produced: check the ADR's status line and the code before acting on a summary. ADR-004 said "*increments 2 and 3 are proposals*"; `GameManager.Loading.cs:174` has materialised a `RemoteSongEntry` before every load since that commit.

But it was unreachable, which is the part worth having gone looking for. The catalogue merge sat behind `YARG_REMOTE_CATALOG=1` — an environment variable with no setting, no documentation, and exactly one occurrence across both repositories: the check itself. A player could not switch it on from inside the game, and nothing told them it existed.

The gate's stated reason was made false by its own commit. Its remarks read "*it is not finished: the entries browse, but they cannot yet be PLAYED — `RemoteSongEntry` returns null from every loader*". True when drafted; false by the time it landed, because `36ca1f2a` is also the commit that added the fetch-on-play step. A gate justified by a limitation its own commit removed is not a feature flag, it is code shipped switched off and then forgotten. That is a distinct failure from the three stale status lines — here the stale sentence was *inside the code it described*, where nobody thinks to distrust it.

Measured before the gate came off, not argued. `Editor/RemotePlaybackProbe.cs` against the live 4,085-song server: three songs known only from the catalogue downloaded, were scanned by the real scanner, and returned real charts.

Song	Cold	Chart	Warm
Fredrik Thordendal's Special Defects — "Solo" Niger Within	1208 ms	158 ms, res 192, 40 sections	9 ms
Sportfreunde Stiller — '54, '74, '90, 2010	131 ms	28 ms, res 480, 9 sections	5 ms
Sportfreunde Stiller — '54,'74,'90,2010	598 ms	68 ms, res 480, 9 sections	4 ms

The probe's decisive check is `LoadChart() != null`, and that choice is the reason the result means anything: "*a file downloaded*" and "*a scan returned something*" both pass for a song that cannot be played.

So the variable became a `ToggleSetting`, **Browse Server Library**, off by default, under *Sync On Startup* in the Song Server tab. The two are alternatives rather than steps — mirroring trades disk for certainty, browsing trades certainty for a library the size of the server's.

`SettingsRowProbe` was watched going red first: with the `en-US.json` strings removed it reported exactly two failures, both naming the new row's missing `Name` and `Description`, and nothing else. That probe exists because **a setting with no entry in `en-US.json` fails nowhere else** — the menu simply renders a blank label.

A correction to the first draft of the setting's own description, caught by reading the code it describes. It said the feature "*uses almost no disk*". `RemoteSongFetcher` has no eviction and nothing else prunes `ServerCache`, so every song ever played from the server is kept for good: it starts using no disk and grows with what is played. The description now says so. **An actual budget or eviction policy is not built**, and with browsing now reachable against a 4,085-song server that is the obvious next thing to want.

The fetch cache got a ceiling — 2026-11-03 (`yarg` `7551cac2`)

Making browsing reachable the previous evening exposed its own gap: every song played from the server was kept and nothing removed it, so the cache grew one song at a time toward the size of the whole server. That is the outcome browsing exists to avoid, reached slowly.

`ServerCacheBudgetGb` bounds it — **10 GB by default, 0 for no limit** — evicting the songs played longest ago first.

Why this deletes when `SongServerSync` deliberately never does. That refusal is about the MIRROR: a library somebody chose to download, possibly the only copy, whose comment says getting it wrong *"costs somebody their library"*. The cache is a different folder holding whatever happened to get played, each item fetchable again. The sweep touches only `ServerCache` — never `ServerLibrary`, never a folder the player added.

The honest difference from the server's own pack cache, which this otherwise copies. There, eviction costs CPU and never data, because an evicted archive is re-packed from a folder still on the same disk. Here the source is another machine: if the server has dropped the song, an evicted copy was the last one. Hence a generous default rather than a tight one, and hence two explicit protections — the song being played, and the song just fetched. The second is the hazard the pack cache had to add a `Vanished` counter for after measuring it happen.

Access order is tracked rather than read off the filesystem, and the reason is ours not Windows'. Last-access updates are enabled on ENG-1 and were measured moving. It would not matter if they were not: `SongContainer` adds the cache to the scanned directories, so a library refresh **reads every `.sng` in it and resets every atime to the same instant**. Our own scanner destroys the ordering signal on every scan. `RecordPlayed` keeps the order in one line-per-song file at the cache root, outside the song folders so no scanner meets it, and a lost index costs eviction quality rather than data — unrecorded songs sort oldest, which is also the right answer for a song fetched and never played.

Red three ways, prediction written down before each run, all three matched.

Break	Predicted	Measured
evict newest first (`OrderByDescending`)	5	FAIL (5), the ordering and size checks
ignore the protected set	3	FAIL (3), both protection checks and the give-up case
never decrement the running total	6	FAIL (6), every "stops when it fits" check

The decision lives in `ChooseEvictions`, pure and public, so it can be measured with no disk, no server and no game — the same shape the setlist guard took the day before. The probe then drives the **real sweep against the real cache folder** on planted songs, because a correct chooser and a sweep that deletes the wrong folder would both pass every check above it.

A probe reported PASS while its own file would not compile

Mid-edit, `ServerCacheBudgetProbe.cs` called a method that did not exist yet. Unity kept the **previously compiled assembly**, the probe ran against that, and it printed `PROBE RESULT: PASS` — for code that was not in the build. The runner printed `COUNT 3` above it, in a section nobody had any reason to read on a passing run.

This is a false green of a kind none of the existing rules catch: the probe is honest, the assertion is sound, and the thing under test is simply not there. **A compile error makes every result below it meaningless**, so the runner now says exactly that where the count used to be, and a green is only read after the error line says `none`.

A full probe sweep, and the harness was wrong six different ways — 2026-11-03

Jay was at work and offered the machine for anything that could run silently, so every headless probe in the fork was run. **Ten of seventeen passed on the first attempt. Not one of the other seven turned out to be a defect in the code** — every one was the harness, the environment, or the way the result was read. That ratio is the finding; the probes themselves were fine.

What the sweep got wrong, in the order it got it wrong:

1. **Six probes live in namespace `Editor`, not `YARG.Editor`.** The sweep assumed one namespace, Unity answered "*executeMethod class could not be found*", and the harness reported `NO RESULT LINE` — **indistinguishable from a probe that ran and printed nothing.** A missing class should never read the same as a silent pass.
2. **A bare-verdict regex of `^(PASS|FAIL)` matched `Failed to load:`**, because `FAIL` prefixes `Failed`. Three probes were reported as having returned a Unity asset warning as their verdict.
3. **The log parser mixed the killed sweep's logs with the re-run's.** The first sweep was stopped mid-flight; its truncated logs sat beside the new ones, and `HostileServerProbe` was reported `FAIL (7)` from a run that had been *killed*. The parser takes the newest log per probe now.
4. **Two probes exited at log line 41 with return code 1** — Unity never started, because the previous instance still held the project lock. The sweep moved on three seconds after seeing a result line, which is not the same as the process having exited. Waiting for a result is not waiting for a lock.
5. **`ScanFolderProbe` wanted `YARG_SCAN_FOLDER` and said so**, in its own output, which nothing was reading.
6. **Two probes share one temp folder and want different things in it.** `HostileServerProbe` needs `%TEMP%\yarg-song-server-smoketest` to hold MIRROR files — named for their chart hash — and says so: "*run Editor.SongServerSyncSmokeTest.Run first*". Yesterday's corpus work left it full of `01-plain.sng`-style packs instead, so the three "known-good" hashes it takes from filenames were not chart hashes at all.

That last one is worth the arithmetic, because the failure was legible and nobody read it. The probe reported `rejected=9` where it expected 6: the six deliberately-bad names it serves, **plus the three it had derived from filenames.** `SongServerSync` refuses anything that is not `^[0-9a-f]{40}$`, correctly, so `01-plain` was refused exactly as a hostile server's garbage would be. Every downstream assertion then failed, including one that computed "*the client accepted -3 of 6 names*" — a negative count, which is the kind of number that should stop a reader cold.

`SongServerSyncSmokeTest` was deliberately excluded from the sweep, and that is why the folder was never repopulated. It calls `SongServerSync.Sync`, which mirrors an ENTIRE library; running it against vault2 would have pulled all 4,085 songs — the same mistake as the night before. It reads `YARG_SONG_SERVER`, a different variable from the single-song `YARG_SERVER_URL` the fetch probes use, and that distinction is the only thing that kept the sweep from downloading the library.

Every one of the seven resolved to a harness or environment problem, and the probes were right. After fixing each cause:

Probe	Result
HostileServerProbe	**PASS**, once the folder held the mirror files it documents needing
RemotePlaybackProbe	**PASS** — it had never started; Unity was still holding the project lock
StartupReachabilityProbe	**PASS**, same cause; it gives up on an unreachable server in 5.1 s
SelfUpdateSmokeTest	**PASS** (`[SMOKE] RESULT: PASS`) — the earlier "no output" was a log read while it was still being written
HttpListenerProbe	**PASS** — a third output format, `[HttpListenerProbe] PASS: …`
ScanFolderProbe	**INCONCLUSIVE, correctly** — no audio backend, so it refuses to claim a verdict rather than reporting the harness's own limits as YARG's

Probe	Result
RemoteQueueSecurityProbe	**informational by design** — it prints measurements, not a verdict. 403 on oversized and cross-origin bodies, 404 on traversal, board capped at 40, identity from the socket

Final tally: 15 pass, 1 inconclusive by design, 1 informational by design. No defects.

Two of those deserve keeping as rules rather than anecdotes:

- `SelfUpdateSmokeTest` **reports only through** `Console.WriteLine`, which goes to stdout and not to `-logFile`. That was survivable while a console window existed; hiding the window — necessary, because it was stealing focus — would have made its output vanish entirely. Its verdict also lives in `EditorApplication.Exit`, which a WMI-created process does not hand back. A runner for these has to capture **stdout and the exit code**, not just the log.
- **Waiting for a result line is not waiting for the lock.** The sweep moved on three seconds after a probe printed its verdict, and the next Unity then failed to open the project because the previous one had not exited. Wait for the process.

What a clean sweep is worth, stated honestly: it says the probes agree with the code. It says nothing about the twelve upstream commits merged on 11-02, which remain unplayed, and nothing about the cache budget in a real game — the budget cannot run headless at all, because `SettingsManager.Settings` needs an audio backend to exist.

The merge's real defect, found by playing: a sort that touches the filesystem — 2026-11-03

Jay turned on *Browse Server Library* and reported four things, which looked like four problems: the song list did not change; he therefore could not play a server song; the badge seemed to appear only on "overlaps"; and the cache limit "doesn't seem to do anything". **All four were one bug, and it is the first real defect the 11-02 merge introduced.**

The log from the run that was supposed to work says both halves of it, four lines apart:

```
SongContainer.cs:MergeRemoteCatalog:211] Song server catalogue: 3952 entries merged, 133 already local
IOException: The filename, directory name, or volume label syntax is incorrect :
  'C:\dev\YARG - Open Source Contributions\yarg\server:\c9c1b767a223...'
  at System.IO.File.GetLastWriteTime (System.String fullPath)
  at YARG.Core.Song.SongEntry.GetLastModifiedTime ()
  at YARG.Menu.MusicLibrary.SongSorting.SortByDateAdded (SongCache, SortedSongs)
  at ...Parallel+<>c__DisplayClass19_0`1[TLocal].<ForWorker>b__1
Rethrow as AggregateException: One or more errors occurred.
```

The merge worked perfectly and the library still showed 128 songs. `RemoteSongEntry` exists precisely so a song can be browsed with no bytes on this disk, so its `ActualLocation` is the synthetic `server://<chart hash>`. `SongEntry.GetLastModifiedTime()` is `File.GetLastWriteTime(ActualLocation)`; Windows resolved `server://…` against the working directory, got `…\yarg\server:\c9c1b…`, and threw. Inside a `Parallel.For` that became an `AggregateException`, which killed `SongSorting.SortEntries`, so `FillContainers` never ran and the browse list was silently discarded.

Neither side is at fault on its own, which is what makes it a merge defect. Our `RemoteSongEntry` had overridden `GetLastWriteTime` since it was written. It had no reason to override `GetLastModifiedTime`, because until 11-02 **nothing called it**. Upstream's #1721 added *Sort by Date Added*, which calls it for every entry in the library. Two correct

changes, merged cleanly, compiled green, and the feature stopped working.

Fixed in `yarg` by overriding it the way the sibling is already overridden:

```
public override DateTime GetLastModifiedTime() => _materialised?.GetLastModifiedTime() ?? _lastWrite;
```

Why nothing caught it

- **The compile could not.** Both sides are type-correct; the base method is `virtual` and has a working default for every entry that has a file.
- **No probe could, as the probes are written.** `RemoteEntryProbe` and `RemotePlaybackProbe` build `RemoteSongEntry` objects and ask whether they browse and play. Neither **sorts a merged library**, which is the only thing that calls the method. `SetlistGuardProbe` and `ServerCacheBudgetProbe` are further away still.
- **The 17-probe sweep on the same day was 15 pass, 1 inconclusive, 1 informational.** A clean sweep says the probes agree with the code; it does not say the probes ask the right questions. This is the concrete instance of the caveat that entry already carried.
- **The headless build could not**, because `SettingsManager.Settings` needs an audio backend, so `BrowseServerLibrary` can never even be read in batchmode.

Three smaller things the same session turned up

- **A toggle that only takes effect on a rescan.** `MergeRemoteCatalog` runs inside `SongContainer.RunRefresh`, so turning *Browse Server Library* on does nothing until the next library scan — a restart, or Settings → Song Manager → refresh. Jay's first report, "doesn't change the quickplay song list", was partly this and partly the crash above. The setting needs an `onChange` that triggers a real scan; note that `RefreshSongs()`, which the other library-affecting settings use, is NOT enough — it calls `RequestContainerRefresh`, which only re-sorts what is already scanned and never reaches the merge.
- **The cache limit appearing to do nothing was a consequence, not a bug.** With no remote entry ever reaching the library, nothing is fetched into `ServerCache`, so there is nothing for the budget to evict. It governs fetched songs only — never the mirror.
- **The badge looked wrong and was right.** With `SongFolders` empty, every song on the machine came from the server, so every song was badged and the badge distinguished nothing. It is only meaningful in a mixed library.

Fixed and measured

`yarg` `5d9b2251`, one line, mirroring the override that was already beside it:

```
public override DateTime GetLastModifiedTime() => _materialised?.GetLastModifiedTime() ?? _lastWrite;
```

Rebuilt and run against the live 4,085-song server:

	before	after
catalogue merged	`3952 entries merged, 133 already local`	same
library the game shows	**128**	**4,080**

	before	after
`IOException` / `AggregateException`	present, killed the sort	**gone**
scan time	0.12 s	2.81 s (the catalogue fetch and merge)
game survives startup	exited	runs

A gap of five is not explained. 3,952 merged plus 133 already local is 4,085, and the library reports 4,080. No `badsongs.txt` was written, so the scanner refused nothing. It may be duplicate chart hashes collapsing, or a filter; it has not been measured and is recorded here as open rather than guessed at. It is 0.12% and did not block anything.

What this costs to believe. The fix is verified by behaviour, not by inspection: a symbol search cannot tell an override from an inherited method, because the base class puts the name in the metadata either way — the same shape as the property trap recorded in `BUILD.md` the same afternoon. The evidence is the library going from 128 to 4,080 with the exception gone, which is the thing anyone actually cares about.

Blockers

None, and no open defects.

> **This section said otherwise until 2026-11-02, and it was wrong.** It described the start-up index as "*designed but not implemented, awaiting Jay's decision*" when the fix had been implemented, deployed and measured in production. Asked what to work on next, this document was read, that line was believed, and the work was picked up — on a defect that no longer existed. The ADR's own status line ("*Accepted and implemented*") is what caught it, within a minute, by contradicting this page.
>
> **It was the second time in one day.** The per-song server badge was offered to Jay the same afternoon from a ROADMAP paragraph that was three days stale, contradicted by the commit message of the work that closed it. Two misfires from the same cause is a pattern, not a slip, and the cause is that a status sentence here duplicates a fact that lives somewhere else — so it can only ever rot. The rule that follows from it is now in this repo's `CLAUDE.md`: **a status line in this document is a summary, never the authority. Check the ADR's status line and the code before acting on it.**

The defect this section used to describe is FIXED, DEPLOYED and MEASURED. Kept here rather than deleted, because the problem is worth understanding and because "there are no blockers" is a claim that should show its work:

The start-up index WAS a data-loss risk, not just an outage. The listener binds only after the whole library is indexed, so a deploy is hard down for the duration — measured on the `0c6fa84` deploy as **10m47s** at 3,067 songs, `/healthz` connection-refused rather than 503, against 3m39s at 1,369. Per-song cost ROSE over that range (0.160 s → 0.209 s), so linear extrapolation understates the next doubling; three points across two library mixes cannot say whether that is superlinear in count or a shift toward loose folders. The danger is not the wait: `POST /api/v1/have` answering `200` with an incomplete index would tell a sync client it has songs the server does not, and a pruning client would delete a player's library. The shipped client already fails closed on any non-200 (`postHave` runs first in `Run` and returns

on a non-200 before downloading or pruning), so **only the server half had to ship** — and the distinction between "indexing" and "genuinely empty" lives in the STATUS CODE, because an empty `200` is a legitimate instruction to prune everything.

What shipped, in `b38df86` "*This change retires 'the port is open' as a readiness signal*": the listener binds before indexing, `/healthz` answers `indexing`, and every library endpoint answers **503 with** `Retry-After: 5` until the index is published. Deployed in `6cbea44` and carried by every build since, including the live `01e0503`. **The outage went from 16m53s to 0.977s** — recorded in `2129210`. Design, measurements and the corrected instrument are in `ADR-008`, whose first out-of-scope follow-up (hashing every file twice) is also now done, at 1.83× with `package_hash` unmoved across all 4,085 songs.

(A "the Unity editor will not start on ENG-1" blocker was filed here on 2026-09-10 and withdrawn the same hour. It was real in the sense that five launches failed, and wrong in every other sense: the cause was not Windows Defender, it was that a process started from a Cowork bridge call inherits a job object which kills Unity's Package Manager child. Launch with `Win32_Process.Create` and it works. That was already written down in the project-state doc, in almost those words, and was not read first. Kept here as a dated retraction rather than deleted, because "a blocker survives on being repeated, not on being true" is already lesson 9 below and this is the same mistake made forward instead of backward.)

Toolchain and credentials, as measured

The Coffeehedake GitHub PAT was rotated on 2026-09-05 and verified (`login=Coffeehedake`, scopes `repo`, `workflow`, `project`, `write:packages`, `delete:packages`, `audit_log`).

Go 1.27.0 was installed on ENG-1 on 2026-09-05, so the toolchain claim is now measured on the machine the work actually happens on: `gofmt` clean, `go vet` exit 0, `go test ./...` green, and all six release targets building — linux/amd64, linux/arm64, linux/armv7, darwin/amd64, darwin/arm64, windows/amd64.

It is a **per-user** install at `%LOCALAPPDATA%\Programs\go`, from the official zip with its SHA-256 verified against `go.dev/dl/?mode=json` before extraction, with `go\bin` and `%USERPROFILE%\go\bin` added to the user `Path`. Per-user rather than machine-wide on purpose: the MSI needs elevation, and an unattended `msiexec /qn` from a non-elevated session fails *silently* — the same failure mode that produced the Bambu Studio update loop. Nothing about this install needs elevation, and `GOPATH` / `GOCACHE` land in the user profile rather than in the projects root — which mattered when that root was a Syncthing folder, and now simply keeps a multi-gigabyte build cache out of a directory full of repos. Promote it to a machine-wide install later if you want; nothing depends on where it lives.

mingw-w64 followed, for the race detector. `go test -race` needs cgo, and cgo needs a C compiler; without one ENG-1 reported `CGO_ENABLED=0` and `-race` failed outright with "requires cgo" rather than passing quietly. WinLibs GCC **16.2.0** (UCRT, POSIX threads, SEH) is now installed the same way — per-user at `%LOCALAPPDATA%\Programs\mingw64`, SHA-256 verified against the publisher's own `.sha256` before extraction, `bin` on the user `Path`. `go env CGO_ENABLED` now reports 1 and the suite is race-clean in about 16 seconds on the first run.

The extraction ran through a **scheduled task** rather than `Start-Process`, because a long child process started from a Cowork session dies with the process tree when a bridge call times out — that is what silently cancelled an earlier `winget` attempt mid-download. A scheduled task is detached from that tree and survives.

And the detector was proved able to go **red** before its green was believed: a throwaway module with eight goroutines incrementing a shared int returned `WARNING: DATA RACE` and exit 1.

Sequencing note

Phases 1 and 2 are the whole of the "#1 priority". Nothing in phases 3–5 should start before a stock YARG client is reading songs from a self-hosted server, because every later decision depends on what that turns out to actually require.